A SHARP RANK-r PARTIAL-TRACE INEQUALITY VIA DOUBLE-SKEW SINGULAR-VALUE BOUNDS

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Abstract. For a completely positive map $\Phi : \mathcal{L}(H) \to \mathcal{L}(H)$ admitting a Kraus representation with $A_a^T = A_a$, we show that a single rank-two bound on its Choi operator generates an r-positive map for every r. We call this *pair amplification.* Specifically, attach $\beta_2(\Phi) = \frac{1}{2}\|J(\Phi)\|_{S(2)}$, the rank-two operator norm of the Choi operator. Then $\Phi \circ \tau + (r-1)\beta_2(\Phi)\big(\Delta - \frac{1}{r}\mathrm{id}\big)$ is r-positive for every r, where $\Delta(X) = \mathrm{Tr}(X)I$ and τ is transposition. Two mechanisms drive it: an ancilla construction and a partial transpose reduce the assertion to pair-indexed antisymmetric modes, over which a complete-graph summation costs only the vertex degree $r-1$; and transpose symmetry — a $\mathbb{Z}/2$-degree condition on the Kraus space — protects one distinguished direction, which sharpens the trace coefficient from 1 to $1/r$.

Its principal corollary, at $\Phi = \Lambda_U \otimes \Lambda_V$ together with a contraction identity for that map, is the arbitrary-matrix rank-r partial-trace inequality: for every operator C of rank at most r on a finite-dimensional bipartite space,

$$\|\mathrm{Tr}_U\, C\|_2^2 + \|\mathrm{Tr}_V\, C\|_2^2 \le r\|C\|_2^2 + \frac{1}{r}|\mathrm{Tr}\, C|^2.$$

Both coefficients are optimal — the first outright, the second once the first is fixed — whenever one factor has dimension at least r and the other at least 2, and for nonzero C equality forces $\operatorname{rank} C = r$. This is the extension beyond normal matrices that Costa Rico and Wolf identify as open. When both local dimensions are at least two, the sharp value $\beta_2 = 1$ is a two-vector action bound on the double-skew space $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$, proved here from the Autonne–Takagi factorization and equivalent to Fu, Gao, and Park's sharp Schmidt-rank-two projection estimate.

The pair-amplification theorem, the main partial-trace theorem, and their dependency chain are machine-checked in Lean 4 end to end. What is certified is the deduction, not the faithfulness of the formal definitions to their informal counterparts; Appendix A gives the repository, the versions, the reported axioms, and the formal–informal correspondence.

Consequences include the exact rank-constrained two-copy score curve, an exact asymmetric tensor-product score theorem with the resulting r-positivity classification, Kronecker-sum Ky Fan bounds, and Schmidt-number witnesses. The score inequalities, their extremizers and rank-constrained thresholds, and the rank-sensitive estimate used in the Kronecker bound are machine-checked; the remaining operator-theoretic interfaces and applications are tracked explicitly in Appendix A.

AI-generation and verification disclosure. The problem selection, prompting, and iterative direction were supplied by the author. The mathematical arguments, claimed results, applications, and exposition were generated by OpenAI's GPT-5.6 and Claude Opus 5. The Lean development of Appendix A was generated by Claude Fable 5 and Claude Opus 5 under the author's direction. It includes a proof of the Autonne–Takagi factorization, so that the development is proved end to end. What a human checked there is the interface — the definitions and the conventions they encode — while Lean checks the deductions built on it. The author did not independently derive the proofs, and no claim of independent human verification is made. The manuscript has not received independent expert review and should not be relied upon until its arguments have been checked by qualified experts.

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1. Introduction

Costa Rico proved the rank-sensitive partial-trace inequality below for normal operators and related its unrestricted rank-two case to two-copy distillability of Werner states [2]. Costa Rico and Wolf subsequently developed partial-trace relations for general matrices, including singular-value majorization results for Kronecker sums and applications to Schmidt-number witnesses and positive maps, while identifying the extension of the same sharp rank-sensitive bound beyond normal matrices as an open problem [3]. Fu, Gao, and Park recently resolved the rank-two endpoint through a sharp estimate for the $S(2)$-norm of a double-antisymmetric projection [4]. Their estimate supplies the sharp local rank-two input, but not by itself a mechanism for combining such information across an r-dimensional range.

The argument below supplies that mechanism and obtains the arbitrary-matrix inequality for every rank, unconditionally. The rank-two input is Lemma 2.1, a two-vector action bound on the double-skew space $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$, which Section 2 proves from the Autonne–Takagi factorization by the argument of [4, §2]; the reduction across an r-dimensional range is new. An r-dimensional ancilla recording an orthonormal basis of the range of C, a partial transpose, and a completely positive map then represent the potentially negative sector by antisymmetric modes indexed by pairs $1 \leq i < j \leq r$, each controlled by that bound. Grouping the modes by the vertices of the complete graph K_r, rather than estimating its $\binom{r}{2}$ edges independently, costs only the vertex degree $r-1$; a cancellation in the distinguished code direction then improves the coefficient of $|\operatorname{Tr} C|^2$ from 1 to the sharp $1/r$.

That mechanism never looks at the double-skew space, and Section 2.1 states it separately as Theorem 2.4, *pair amplification*. Its inputs are two: a completely positive Φ whose Kraus operators are transpose-symmetric, and the single number $\beta_2(\Phi)$ of Definition 2.3. Theorem 1.2 is the instance $\Phi = \Lambda_U \otimes \Lambda_V$, where $\beta_2 = 1$; Remark 3.3 records what the two obvious further instances need. Beyond threshold statements, the applications determine the exact rank-constrained score curves — the minimum of the normalized two-copy filter form over nonzero matrices of rank at most r, as in Equation (52) — for both symmetric and asymmetric two-copy filters; the asymmetric formula yields a complete r-positivity classification for the corresponding tensor-product maps.

The Werner-state and distillability terminology used in the applications follows the standard conventions [16, 6]. Throughout, $\|\cdot\|_2$ denotes the Hilbert–Schmidt, or Schatten-2, norm. The notation Tr_U means the partial trace *over* the factor U. Inner products are conjugate-linear in the first argument and linear in the second. Relative to chosen orthonormal bases, vectorization is normalized by

$$\operatorname{vec}(|x\rangle\langle y|) = x \otimes \overline{y}.$$

The symbol $\operatorname{SR}(z)$ denotes the Schmidt rank of a vector z, and $\operatorname{SN}(\sigma)$ denotes the Schmidt number of a positive operator or state σ, always across the bipartition indicated in context. With the above normalization, a vector has Schmidt rank at most k across the output–input cut exactly when it is the vectorization of a matrix of rank at most k; that is,

$$\operatorname{SR}\big(\operatorname{vec}(A)\big) = \operatorname{rank} A, \tag{1}$$

a fact used repeatedly below. Finally, for an operator on a twofold tensor power, Tr_1 denotes the partial trace *over* the first local factor, leaving an operator on the second, and Tr_2 traces over the second.

The paper is self-contained modulo one classical ingredient, the Autonne–Takagi factorization of a complex symmetric matrix [5, Cor. 4.4.4(c)], from which Lemma 2.1 is proved in Section 2 following the strategy of [4, §2]; so Lemma 2.1 is not imported from their work but proved, and the formal development of Appendix A proves the Autonne–Takagi factorization as well, making that chain complete end to end. Remark 2.2 records that Lemma 2.1 is *equivalent* to the sharp projection estimate of Fu, Gao, and Park, so their Theorem 2.4 may be substituted for the proof given here, and conversely.

The main theorem is the mechanism. For a completely positive Φ on $\mathcal{L}(H)$ with Choi operator $J(\Phi) = \sum_a |\mathrm{vec}\, A_a\rangle\langle \mathrm{vec}\, A_a|$, write $\beta_2(\Phi) := \frac{1}{2}\|J(\Phi)\|_{S(2)}$ for half the supremum of $\langle z, J(\Phi)z\rangle$ over unit vectors of Schmidt rank at most two (Definition 2.3), and let $\Delta(X) := \mathrm{Tr}(X)I$ and τ be transposition.

Theorem 1.1 (Pair amplification). *Let Φ be completely positive on $\mathcal{L}(H)$, $\dim H < \infty$, admitting a Kraus representation with $A_a^T = A_a$. Then for every $r \geq 1$ the map*

$$\Psi_{r,\Phi} := \Phi \circ \tau + (r-1)\beta_2(\Phi)\left(\Delta - \tfrac{1}{r}\mathrm{id}\right)$$

is r-positive.

One number, computed at level two, controls every level r; the cost grows linearly, with slope $\beta_2(\Phi)$, and vanishes at $r = 1$, where the assertion is that $\Phi \circ \tau$ is positive. The two hypotheses act separately, and Theorem 2.4 states the operator forms this splits into: complete positivity and β_2 give the bound, and transpose symmetry — a $\mathbb{Z}/2$-degree condition on the Kraus space, Remark 2.6 — supplies the $\frac{1}{r}\mathrm{id}$, which is exactly the difference between the coefficients 1 and $1/r$ below. Section 2.1 proves it.

Its principal corollary is the estimate the introduction began with. It is not a special case: passing from r-positivity to the marginals needs a contraction identity for the particular map $\Lambda_U \otimes \Lambda_V$ (Equation (23)), and Remark 3.2 shows that β_2 alone cannot carry it.

Theorem 1.2 (Rank-r partial-trace inequality). *Let U and V be finite-dimensional complex Hilbert spaces. If $C \in \mathcal{L}(U \otimes V)$ is nonzero and $r_0 := \mathrm{rank}\, C$, then*

$$\|\mathrm{Tr}_U\, C\|_2^2 + \|\mathrm{Tr}_V\, C\|_2^2 \leq r_0\|C\|_2^2 + \frac{1}{r_0}|\mathrm{Tr}\, C|^2. \tag{2}$$

Consequently, for every $r \in \mathbb{N}_{\geq 1}$ such that $\mathrm{rank}\, C \leq r$,

$$\|\mathrm{Tr}_U\, C\|_2^2 + \|\mathrm{Tr}_V\, C\|_2^2 \leq r\|C\|_2^2 + \frac{1}{r}|\mathrm{Tr}\, C|^2. \tag{3}$$

The assertion is immediate when $C = 0$.

Equation (3) was obtained independently and concurrently by Fraser, Huber, Pozsgay and Vona [8], by a different route. Song and Chen [9] and Bharti, Gajjala and Haug [10] obtain the rank-two case; Song and Chen also give a two-parameter extension and an activation result, while Bharti, Gajjala and Haug give finite-copy extensions. All four accounts resolve the two-copy distillability question through Costa Rico's equivalence. The overlap is the inequality and its rank-two consequences; what is particular to this account is the reusable derivation from Theorem 1.1, together with the formal development of Appendix A.

Remark 1.3 (Optimality of the coefficients). The coefficients are optimal whenever $\dim U \geq r$ and $\dim V \geq 2$, or with U and V interchanged. More precisely, the coefficient r of $\|C\|_2^2$ is optimal, and conditional on choosing this optimal coefficient, the coefficient $1/r$ of $|\mathrm{Tr}\, C|^2$ is optimal.

Indeed, choose

$$C = P_r \otimes |v\rangle\langle w|,$$

where P_r is a rank-r projection on U and $v, w \in V$ are unit vectors. Then

$$\begin{aligned}
\|\mathrm{Tr}_U\, C\|_2^2 &= r^2, \\
\|\mathrm{Tr}_V\, C\|_2^2 &= r|\langle w, v\rangle|^2, \\
\|C\|_2^2 &= r, \\
|\mathrm{Tr}\, C|^2 &= r^2|\langle w, v\rangle|^2.
\end{aligned}$$

Thus equality holds in Equation (3), and the displayed extremizer has $\operatorname{rank} C = r$ exactly. More generally, for nonzero C equality in the rank-r inequality requires $\operatorname{rank} C = r$; the passage from the exact-rank inequality to a strictly larger rank parameter is strict. This is why the sharpness example is built from a rank-r projection. For a bound of the form

$$\|\mathrm{Tr}_U C\|_2^2 + \|\mathrm{Tr}_V C\|_2^2 \le a\|C\|_2^2 + b|\mathrm{Tr}\, C|^2,$$

taking $v \perp w$ forces $a \ge r$. Once $a = r$ is fixed, taking $v = w$ forces $b \ge 1/r$.

The proof of Theorem 1.2, and of every result it depends on, has been formalized in Lean 4, as have the statements of Remark 1.3 and Lemma 4.7, the algebraic score applications, and the rank-sensitive core of Corollary 4.11. Appendix A records the repository, the toolchain and Mathlib pins, the reported axioms, and the correspondence between the formal and informal statements.

2. Double-Skew Estimates and the Marginal Operator Inequality

We first prove the double-skew action bound used below. The route is that of [4, §2], with the Autonne–Takagi factorization in place of their Theorem 2.4; Remark 2.2 records that the two are equivalent.

Lemma 2.1 (Double-skew action bound)**.** *Fix orthonormal bases of U and V, and let*

$$\mathfrak{so}(U) := \{L \in \mathcal{L}(U) : L^T = -L\}, \qquad \mathfrak{so}(V) := \{M \in \mathcal{L}(V) : M^T = -M\}.$$

For every K in the linear span of

$$\{L \otimes M : L \in \mathfrak{so}(U),\ M \in \mathfrak{so}(V)\},$$

and every orthonormal pair $x, y \in U \otimes V$,

$$\|Kx\|^2 + \|Ky\|^2 \le \frac{1}{2}\|K\|_2^2. \tag{4}$$

Proof. Index operators on $U \otimes V$ by pairs relative to the fixed bases, writing $N_{(a,b),(a',b')}$, and define the two partial transposes

$$(N^{T_U})_{(a,b),(a',b')} := N_{(a',b),(a,b')}, \qquad (N^{T_V})_{(a,b),(a',b')} := N_{(a,b'),(a',b)},$$

whose composite is the ordinary transpose. Both are self-adjoint involutions for the Hilbert–Schmidt inner product and they commute, so

$$\Pi := \tfrac{1}{2}(1 - T_U)\, \tfrac{1}{2}(1 - T_V)$$

is the orthogonal projection of $\mathcal{L}(U \otimes V)$ onto $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$. Hence $\Pi K = K$; and since a tensor product of two skew-symmetric matrices is symmetric, also $K^T = K$.

Step 1: a symmetric Schmidt decomposition. Because $K^T = K$, the Autonne–Takagi factorization [5, Cor. 4.4.4(c)] supplies an orthonormal basis $w_1, \dots, w_n$ of $U \otimes V$ and reals $s_i \ge 0$ with

$$K = \sum_i s_i\, w_i w_i^T. \tag{5}$$

The $w_i w_i^T$ are themselves orthonormal in the Hilbert–Schmidt inner product, because $\langle uu^T, vv^T\rangle_{\mathrm{HS}} = \langle u, v\rangle^2$.

Step 2: the two-term bound. Write M_u for the $\dim U \times \dim V$ coefficient matrix of $u \in U \otimes V$, so that $\|M_u\|_2 = \|u\|$. Carrying out the two inner sums separately gives the flip trace identity

$$\langle uu^T, (vv^T)^{T_U}\rangle_{\mathrm{HS}} = \mathrm{Tr}\big((M_u^* M_v)^2\big). \tag{6}$$

Let $N := s_1 w_1 w_1^T + s_2 w_2 w_2^T$ with $s_1, s_2 \ge 0$, write $M_i := M_{w_i}$ and $c_i := \|M_i^* M_i\|_2$. Expanding by Equation (6), using that $M_i^* M_i$ is self-adjoint for the diagonal terms and that the two cross terms are complex conjugates,

$$\langle N, N^{T_U}\rangle_{\mathrm{HS}} = s_1^2 c_1^2 + 2 s_1 s_2\, \mathrm{Re}\,\mathrm{Tr}\big((M_1^* M_2)^2\big) + s_2^2 c_2^2.$$

Two applications of Cauchy–Schwarz for $\langle\cdot,\cdot\rangle_{\mathrm{HS}}$, namely

$$|\mathrm{Tr}(X^2)| = |\langle X^*, X\rangle_{\mathrm{HS}}| \le \|X\|_2^2, \qquad \|A^*B\|_2^2 = \langle AA^*, BB^*\rangle_{\mathrm{HS}} \le \|A^*A\|_2\,\|B^*B\|_2,$$

the second using $\|AA^*\|_2 = \|A^*A\|_2$, bound the cross term below by $-2s_1s_2c_1c_2$. Therefore

$$\langle N, N^{T_U}\rangle_{\mathrm{HS}} \ge (s_1c_1 - s_2c_2)^2 \ge 0. \tag{7}$$

Since $N^T = N$ forces $N^{T_V} = N^{T_U}$, expanding Π leaves only two distinct terms, and Equation (7) gives

$$\langle N, \Pi N\rangle_{\mathrm{HS}} = \tfrac{1}{2}(\|N\|_2^2 - \langle N, N^{T_U}\rangle_{\mathrm{HS}}) \le \tfrac{1}{2}\|N\|_2^2. \tag{8}$$

This is the only place the hypothesis $\dim \le 2$ of [4, Lem. 2.3] is used, and Equation (7) is that lemma.

Step 3: truncation. Fix $a \ne b$ and put $K_{ab} := s_a w_a w_a^T + s_b w_b w_b^T$. Orthonormality of the $w_i w_i^T$ gives $\|K_{ab}\|_2^2 = \langle K, K_{ab}\rangle_{\mathrm{HS}} = s_a^2 + s_b^2 =: A$. As Π is a self-adjoint idempotent with $\Pi K = K$,

$$A = \langle K, K_{ab}\rangle_{\mathrm{HS}} = \langle \Pi K, K_{ab}\rangle_{\mathrm{HS}} = \langle K, \Pi K_{ab}\rangle_{\mathrm{HS}} \le \|K\|_2\,\|\Pi K_{ab}\|_2,$$

while $\|\Pi K_{ab}\|_2^2 = \langle K_{ab}, \Pi K_{ab}\rangle_{\mathrm{HS}} \le \frac{1}{2}A$ by Equation (8). Squaring the first display and substituting the second yields $A^2 \le \frac{1}{2}A\|K\|_2^2$. If $A = 0$ there is nothing to prove; otherwise one factor of A cancels, and

$$s_a^2 + s_b^2 \le \tfrac{1}{2}\|K\|_2^2 \qquad \text{for } \textit{every} \text{ pair } a \ne b. \tag{9}$$

Step 4: from pairs to the action. By Equation (5) and orthonormality, $K^*K = \sum_i s_i^2 |\overline{w_i}\rangle\langle\overline{w_i}|$, so for an orthonormal pair x, y,

$$\|Kx\|^2 + \|Ky\|^2 = \sum_i s_i^2\theta_i, \qquad \theta_i := |\langle \overline{w_i}, x\rangle|^2 + |\langle \overline{w_i}, y\rangle|^2.$$

Each θ_i equals $\|P\overline{w_i}\|^2$ for the rank-two orthogonal projection P onto $\mathrm{span}\{x, y\}$, so $\theta_i \in [0,1]$; and $\sum_i \theta_i = \mathrm{Tr}\,P = 2$ because $\{\overline{w_i}\}$ is an orthonormal basis. For weights of this kind and any $c_i \ge 0$, letting a, b index the two largest c_i and bounding every remaining term by $c_b\theta_i$,

$$\sum_i c_i\theta_i \le c_a\theta_a + c_b(2 - \theta_a) = 2c_b + \theta_a(c_a - c_b) \le c_a + c_b.$$

Applying this with $c_i = s_i^2$, and then Equation (9) to that pair, gives Equation (4). □

Remark 2.2 (Equivalence with the Fu–Gao–Park estimate). Lemma 2.1 is equivalent to [4, Thm. 2.4], so that estimate may be substituted for the proof above.

Label the four vectorization factors $U_{\mathrm{out}}, V_{\mathrm{out}}, U_{\mathrm{in}}, V_{\mathrm{in}}$ as $1, 2, 3, 4$; then the antisymmetrized pairs $(U_{\mathrm{out}}, U_{\mathrm{in}})$ and $(V_{\mathrm{out}}, V_{\mathrm{in}})$ are Fu–Gao–Park's $(1,3)$ and $(2,4)$, the cut $(U_{\mathrm{out}}V_{\mathrm{out}}) : (U_{\mathrm{in}}V_{\mathrm{in}})$ is their 12:34, and vec intertwines Π with $Q_- := P^{(1,3)}_{\wedge^2 U} \otimes P^{(2,4)}_{\wedge^2 V}$. Their estimate states that, for every z of Schmidt rank at most two across that cut,

$$\langle z, Q_- z\rangle \le \tfrac{1}{2}\|z\|^2, \tag{10}$$

both sides being homogeneous of degree two, so that no normalization is needed. By Equation (1) this says exactly that $\langle M, \Pi M\rangle_{\mathrm{HS}} \le \frac{1}{2}\|M\|_2^2$ for every $M \in \mathcal{L}(U \otimes V)$ with $\mathrm{rank}\,M \le 2$.

Their estimate implies Lemma 2.1. Let $P = |x\rangle\langle x| + |y\rangle\langle y|$ and $A = \|KP\|_2^2$. Since $P^2 = P = P^*$, $\|Kx\|^2 + \|Ky\|^2 = A$ and $\mathrm{rank}(KP) \le 2$; and as Π is a self-adjoint idempotent fixing K, $A = \langle K, KP\rangle_{\mathrm{HS}} = \langle K, \Pi(KP)\rangle_{\mathrm{HS}} \le \|K\|_2\|\Pi(KP)\|_2$, while $\|\Pi(KP)\|_2^2 \le \frac{1}{2}A$. Cancelling one factor of A gives Equation (4). (A block-zero extension reduces unequal local dimensions to the square case they state, since extending a skew-symmetric matrix by zero preserves skew-symmetry and the Hilbert–Schmidt norm.)

Conversely. Given M with $\mathrm{rank}\,M \le 2$, choose an orthonormal pair v_1, v_2 spanning a subspace that contains the row space $\mathrm{ran}\,M^* = (\ker M)^\perp$, and set $u_i := Mv_i$; then $M = \sum_{i\le 2}|u_i\rangle\langle v_i|$ and $\|M\|_2^2 = \sum_i\|u_i\|^2$. (If $\dim(U \otimes V) = 1$ then $\Pi = 0$ and there is nothing to prove.) Put

$K := \Pi M$. Then $\|K\|_2^2 = \langle K, M\rangle_{\mathrm{HS}} = \sum_i \langle K v_i, u_i\rangle \le \left(\sum_i \|K v_i\|^2\right)^{1/2} \|M\|_2 \le \left(\frac{1}{2}\|K\|_2^2\right)^{1/2} \|M\|_2$ by Equation (4), whence $\langle M, \Pi M\rangle_{\mathrm{HS}} = \|\Pi M\|_2^2 \le \frac{1}{2}\|M\|_2^2$.

Both directions are machine-checked; see Appendix A.

2.1. **Pair amplification.** Lemma 2.1 is one instance of a general mechanism, and it is cleaner to state the general form first and specialize. What the argument of Section 3 consumes is a single number attached to a completely positive map.

Definition 2.3 (Rank-two Choi constant)**.** *Let Φ be completely positive on $\mathcal{L}(H)$ with Kraus family $\{A_a\}$, and let*

$$J(\Phi) := \sum_a |\mathrm{vec}\, A_a\rangle\langle \mathrm{vec}\, A_a|$$

be its Choi operator. Set

$$\beta_2(\Phi) := \tfrac{1}{2}\|J(\Phi)\|_{S(2)} = \tfrac{1}{2}\sup\{\langle z, J(\Phi) z\rangle : \mathrm{SR}(z) \le 2,\ \|z\| = 1\}.$$

$J(\Phi)$ is the same operator for every Kraus family of Φ, so β_2 depends on Φ alone — no minimality and no orthogonality of the family is required. It is positively homogeneous in Φ — and convex, being a supremum of linear functionals, but not additive — so a map and its constant always carry the same scale and no normalization convention has to be tracked.

Writing $K_\zeta := \sum_a \zeta_a A_a$ and s_i for singular values, three descriptions coincide:

$$2\beta_2(\Phi) = \|J(\Phi)\|_{S(2)} = \sup_{\|\zeta\|=1} \left(s_1(K_\zeta)^2 + s_2(K_\zeta)^2\right) = \sup_{\|\zeta\|=1} \left\|K_\zeta K_\zeta^*\right\|_{(2)}, \tag{11}$$

$\|\cdot\|_{(k)}$ being the Ky Fan k-norm; any of the three may serve as the definition. The middle equality is the $S(k)$-norm projection duality of [7] at $k = 2$, and the right-hand quantity is the threshold characterizing k-positivity of $a\,\mathrm{Tr}(\cdot) I - \Phi$ in [13, Thm. 3.2], so $2\beta_2$ is their a_2. By Remark 2.2, $J(\Lambda_U \otimes \Lambda_V)$ is $4Q_-$. If $\dim U, \dim V \ge 2$, then $\|Q_-\|_{S(2)} = \frac{1}{2}$, so

$$\beta_2(\Lambda_U \otimes \Lambda_V) = 1. \tag{12}$$

If either local dimension is below two, the double-skew space and the map vanish, and the sharp value is instead $\beta_2 = 0$. The upper bound $\beta_2 \le 1$, which is all the proof of Theorem 1.2 consumes, is valid in every dimension.

Theorem 2.4 (Pair amplification, operator forms)**.** *Let Φ be completely positive on $\mathcal{L}(U \otimes V)$. Then, in the notation of Proposition 2.5, for every $r \ge 1$ and every orthonormal family $e_1, \dots, e_r$ in $U \otimes V$,*

$$(\Phi \otimes \mathrm{id}_Q)(P_-) \preceq (r-1)\beta_2(\Phi)\, I_{UVQ}, \tag{13}$$

$$(\Phi \otimes \mathrm{id}_Q)(\rho_0^{T_{UV}}) + (r-1)\beta_2(\Phi)\, I_{UVQ} \succeq 0. \tag{14}$$

If moreover the Kraus operators of Φ are transpose-symmetric, *$A_a^T = A_a$, then I_{UVQ} may be replaced by $I_{UVQ} - \Pi_e$ throughout:*

$$(\Phi \otimes \mathrm{id}_Q)(P_-) \preceq (r-1)\beta_2(\Phi)\, (I_{UVQ} - \Pi_e), \tag{15}$$

$$(\Phi \otimes \mathrm{id}_Q)(\rho_0^{T_{UV}}) + (r-1)\beta_2(\Phi)\, (I_{UVQ} - \Pi_e) \succeq 0. \tag{16}$$

The two hypotheses act separately, and the gap between the two pairs of displays is the single direction δ_e. Complete positivity and β_2 give Equations (13) and (14); transpose symmetry, and nothing else, supplies the projection. Remark 2.6 identifies that hypothesis as a $\mathbb{Z}/2$-degree condition and shows what the sharpening is worth: the coefficient $1/r$ of $|\mathrm{Tr}\, C|^2$ in Theorem 1.2 rather than 1.

In the transpose-symmetric case Equation (16) is Theorem 1.1: it says that

$$\Psi_{r,\Phi} := \Phi \circ \tau + (r-1)\beta_2(\Phi) \left(\Delta - \tfrac{1}{r}\mathrm{id}\right)$$

is r-*positive*, meaning that $(\Psi_{r,\Phi} \otimes \mathrm{id}_{\mathbb{C}^r})(\rho) \succeq 0$ for every positive semidefinite ρ on $(U \otimes V) \otimes \mathbb{C}^r$ — equivalently, by the Choi–Jamiołkowski correspondence used throughout Section 4, that its Choi operator has nonnegative expectation on every vector of Schmidt rank at most r. Remark 2.7 carries out that translation and identifies the resulting map.

The proof occupies Section 3; Equation (15) is Equation (25) with β_2 in place of 1, and each of Equations (14) and (16) follows from the display above it because P_+ survives Φ and $P_+ - P_-$ is twice the partial transpose. The constant β_2 enters through one estimate on the Kraus span (Equation (32)), and transpose symmetry through one orthogonality (Equation (34)). Nothing else about Φ is used — in particular nothing about $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$ beyond Equation (12).

At $\Phi = \Lambda_U \otimes \Lambda_V$ and $\beta_2 = 1$, Theorem 2.4 is the operator inequality we now record in the form Section 3 uses.

Proposition 2.5 (Rank-r marginal operator inequality)**.** *Let $Q = \mathbb{C}^r$, let $e_1, \dots, e_r$ be an orthonormal family in $U \otimes V$, and let $q_1, \dots, q_r$ be an orthonormal basis of Q. Define the unnormalized code vector and its rank-one operator by*

$$\delta_e := \sum_{i=1}^{r} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle\delta_e|, \qquad \Pi_e := \frac{1}{r}\rho_0.$$

Thus Π_e is the orthogonal projection onto $\mathrm{span}\{\delta_e\}$*. Write*

$$\rho^0_{UQ} := \mathrm{Tr}_V\, \rho_0, \qquad \rho^0_{VQ} := \mathrm{Tr}_U\, \rho_0.$$

Then

$$H_r^{(0)} := rI_{UVQ} + \Pi_e - \iota_{UQ}(\rho^0_{UQ}) - \iota_{VQ}(\rho^0_{VQ}) \succeq 0. \tag{17}$$

For $S \subseteq \{U, V, Q\}$, the map

$$\iota_S : \mathcal{L}\left(\bigotimes_{X \in S} X\right) \longrightarrow \mathcal{L}(U \otimes V \otimes Q)$$

uses the order inherited from U, V, Q: it tensors with identities on omitted factors and conjugates by the unique permutation unitary that restores the order $U \otimes V \otimes Q$. For example, $\iota_{UQ}(X)$ is the operator that acts as X on the labeled U, Q factors and as the identity on V.

Proof. Fix orthonormal bases of U and V, relative to which transposition and the skew-symmetric operator spaces are defined. Although the proof uses these bases, the resulting operator inequality is basis independent.

Expanding ρ_0 and transposing the UV factor gives

$$\begin{aligned} \rho_0^{T_{UV}} &= \sum_{i,j=1}^{r} \left(|e_i\rangle\langle e_j|\right)^T \otimes |q_i\rangle\langle q_j| \\ &= \sum_{i,j=1}^{r} |\overline{e}_j \otimes q_i\rangle\langle \overline{e}_i \otimes q_j|. \end{aligned} \tag{18}$$

Let

$$S := \mathrm{span}\{\overline{e}_1, \dots, \overline{e}_r\},$$

and let $J : Q \to S$ be the unitary determined by $Jq_i = \overline{e}_i$. Define a self-adjoint unitary F_J on $S \otimes Q$ by

$$F_J(s \otimes q) := Jq \otimes J^{-1}s.$$

In particular,

$$F_J(\overline{e}_i \otimes q_j) = \overline{e}_j \otimes q_i.$$

Thus the sum in Equation (18) is F_J on $S \otimes Q$. Let

$$P_\pm := \frac{1}{2}(I_{S \otimes Q} \pm F_J)$$

be its positive and negative eigenspace projections, extended by zero on $(S \otimes Q)^{\perp}$; thus $P_+ + P_-$ is the orthogonal projection onto $S \otimes Q$, not the identity on $U \otimes V \otimes Q$. Then

$$\rho_0^{T_{UV}} = P_+ - P_-. \tag{19}$$

Explicitly,

$$P_- = \sum_{1 \le i < j \le r} |\eta_{ij}\rangle\langle\eta_{ij}|,$$

where

$$\eta_{ij} = \frac{1}{\sqrt{2}}(\overline{e}_i \otimes q_j - \overline{e}_j \otimes q_i). \tag{20}$$

For a finite-dimensional Hilbert space E, define

$$\Lambda_E(X) := \operatorname{Tr}(X) I_E - X^T.$$

If $L_{ab} = E_{ab} - E_{ba}$, then

$$\Lambda_E(X) = \sum_{a<b} L_{ab} X L_{ab}^*. \tag{21}$$

Thus Λ_E is completely positive. Equivalently, its Choi operator is

$$J(\Lambda_E) = \sum_{i,j} |i\rangle\langle j| \otimes \Lambda_E(E_{ij}) = I - F,$$

whose spectrum is contained in $\{0, 2\}$.

Define

$$\Phi := \Lambda_U \otimes \Lambda_V \quad \text{on } \mathcal{L}(U \otimes V), \qquad \Phi_Q := \Phi \otimes \mathrm{id}_Q.$$

Thus Φ is the map to which β_2 of Definition 2.3 is attached, and Φ_Q is its amplification, the operator that acts on $U \otimes V \otimes Q$. For an operator Y on $U \otimes V \otimes Q$, direct expansion gives

$$\begin{aligned} \Phi_Q(Y) = \iota_Q(\operatorname{Tr}_{UV} Y) - \iota_{VQ}\Big((\operatorname{Tr}_U Y)^{T_V}\Big) \\ - \iota_{UQ}\Big((\operatorname{Tr}_V Y)^{T_U}\Big) + Y^{T_{UV}}. \end{aligned} \tag{22}$$

Applying Equation (22) to $Y = \rho_0^{T_{UV}}$ gives

$$\Phi_Q(\rho_0^{T_{UV}}) = I_{UVQ} - \iota_{UQ}(\rho^0_{UQ}) - \iota_{VQ}(\rho^0_{VQ}) + \rho_0. \tag{23}$$

Using Equation (19), we therefore obtain

$$H_r^{(0)} = \Phi_Q(P_+) + \Big[(r-1)(I_{UVQ} - \Pi_e) - \Phi_Q(P_-)\Big]. \tag{24}$$

Since $\Phi_Q(P_+) \succeq 0$, it remains to prove

$$\Phi_Q(P_-) \preceq (r-1)(I_{UVQ} - \Pi_e). \tag{25}$$

Let $\{L_\alpha\}_{\alpha \in \mathcal{A}_U}$ and $\{M_\beta\}_{\beta \in \mathcal{A}_V}$ be the standard skew-symmetric Kraus operators for Λ_U and Λ_V, normalized by

$$\langle L_\alpha, L_{\alpha'} \rangle_{\mathrm{HS}} = 2\delta_{\alpha\alpha'}, \qquad \langle M_\beta, M_{\beta'} \rangle_{\mathrm{HS}} = 2\delta_{\beta\beta'}.$$

Put

$$\mathcal{I}_r := \{(i, j, \alpha, \beta) : 1 \le i < j \le r,\ \alpha \in \mathcal{A}_U,\ \beta \in \mathcal{A}_V\}.$$

Define $\mathcal{T} : \ell_2(\mathcal{I}_r) \to U \otimes V \otimes Q$ by

$$\mathcal{T} c := \sum_{1 \le i < j \le r} \sum_{\alpha, \beta} c^{(ij)}_{\alpha\beta} (L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}. \tag{26}$$

Thus the column indexed by $(i, j, \alpha, \beta) \in \mathcal{I}_r$ is $(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}$. Using the Kraus form of Φ_Q and the rank-one expansion of P_- therefore gives

$$\Phi_Q(P_-) = \sum_{i<j} \sum_{\alpha,\beta} |(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}\rangle\langle(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}|$$

$$= \mathcal{T}\mathcal{T}^*. \tag{27}$$

For each edge (i,j) of the complete graph K_r, define

$$K_c^{(ij)} := \sum_{\alpha,\beta} c_{\alpha\beta}^{(ij)} L_\alpha \otimes M_\beta.$$

Then $K_c^{(ij)}$ lies in the linear span of

$$\{L \otimes M : L \in \mathfrak{so}(U),\ M \in \mathfrak{so}(V)\},$$

and orthogonality of the Kraus operators gives

$$\begin{aligned}\|K_c^{(ij)}\|_2^2 &= \sum_{\alpha,\beta} |c_{\alpha\beta}^{(ij)}|^2 \|L_\alpha \otimes M_\beta\|_2^2 \\ &= 4\|c^{(ij)}\|_2^2.\end{aligned} \tag{28}$$

Grouping the output of $\mathcal{T}$ according to the basis $\{q_k\}$ gives

$$\mathcal{T}c = \frac{1}{\sqrt{2}} \sum_{k=1}^{r} \left(\sum_{i<k} K_c^{(ik)} \overline{e}_i - \sum_{k<j} K_c^{(kj)} \overline{e}_j \right) \otimes q_k. \tag{29}$$

At each vertex k, there are $r-1$ incident terms. Hence Cauchy–Schwarz yields

$$\begin{aligned}\|\mathcal{T}c\|^2 &\le \frac{r-1}{2} \sum_{k=1}^{r} \left(\sum_{i<k} \|K_c^{(ik)} \overline{e}_i\|^2 + \sum_{k<j} \|K_c^{(kj)} \overline{e}_j\|^2 \right) \\ &= \frac{r-1}{2} \sum_{1 \le i < j \le r} \left(\|K_c^{(ij)} \overline{e}_i\|^2 + \|K_c^{(ij)} \overline{e}_j\|^2 \right).\end{aligned} \tag{30}$$

Each parenthesized expression is bounded by $\beta_2(\Phi)$ directly, without Equation (28) and without any property of $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$. Write $\{A_a\}$ for the Kraus family of Φ on $\mathcal{L}(U \otimes V)$, here $A_{\alpha\beta} = L_\alpha \otimes M_\beta$, and let $S : c \mapsto \sum_a c_a A_a$ be its synthesis operator, so that $SS^* = J(\Phi)$ under vec. For an orthonormal pair x, y and $P := |x\rangle\langle x| + |y\rangle\langle y|$, self-adjoint idempotence of P gives $\langle K, KP\rangle_{\mathrm{HS}} = \|KP\|_2^2$, so $M := KP/\|KP\|_2$ has $\operatorname{rank} M \le 2$, $\|M\|_2 = 1$ and $\langle M, K\rangle_{\mathrm{HS}} = \|KP\|_2$; hence $\|KP\|_2 \le \sup\{|\langle M, K\rangle_{\mathrm{HS}}| : \operatorname{rank} M \le 2,\ \|M\|_2 = 1\}$, with equality only when P compresses onto a dominant pair of singular directions. With $K = Sc$ this suffices:

$$\|Kx\|^2 + \|Ky\|^2 = \|KP\|_2^2 \le \sup_M |\langle S^*M, c\rangle|^2 \le \|c\|_2^2 \sup_M \|S^*M\|_2^2 = \|c\|_2^2\, \|J(\Phi)\|_{S(2)}, \tag{31}$$

the suprema being over $\operatorname{rank} M \le 2$, $\|M\|_2 = 1$. Hence

$$\|Kx\|^2 + \|Ky\|^2 \le 2\beta_2(\Phi)\|c\|_2^2. \tag{32}$$

This is $\|S\|^2 \le \|SS^*\|$ in the $S(2)$-norm, and is the elementary half of the projection duality of [7]; that a rank-two compression suffices to test the pair is [11, Prop. 1.1]. Neither minimality nor orthogonality of $\{A_a\}$ is used — the projection Q_- enters only through the number $\beta_2(\Phi)$, and $\Pi K = K$ is replaced by membership of K in the Kraus span. Substituting Equation (32) into Equation (30),

$$\begin{aligned}\|\mathcal{T}c\|^2 &\le \frac{r-1}{2} \cdot 2\beta_2(\Phi) \sum_{i<j} \|c^{(ij)}\|_2^2 \\ &= (r-1)\beta_2(\Phi)\|c\|_2^2.\end{aligned} \tag{33}$$

For $\Phi = \Lambda_U \otimes \Lambda_V$ this is $(r-1)\|c\|_2^2$ by Equation (12), and the estimate is attained already when $r = 2$ and $\dim U = \dim V = 2$. Then every vertex has degree one and $\mathfrak{so}(2) \otimes \mathfrak{so}(2)$ is one-dimensional; its nonzero elements are scalar multiples of a unitary, so every orthonormal pair saturates Equation (4). There is no slack anywhere in that $r = 2$ chain.

Thus

$$\mathcal{T}\mathcal{T}^* \preceq (r-1)\beta_2(\Phi)\, I_{UVQ}.$$

It remains to use the orthogonality of the range of $\mathcal{T}$ to δ_e. Since each tensor product of two skew-symmetric operators is symmetric,

$$K^T = K \qquad \text{for every } K \text{ in the above linear span.}$$

Consequently,

$$\begin{aligned}\langle \delta_e, \mathcal{T}c\rangle &= \frac{1}{\sqrt{2}} \sum_{i<j} \left(\overline{e}_j^T K_c^{(ij)} \overline{e}_i - \overline{e}_i^T K_c^{(ij)} \overline{e}_j \right) \\ &= 0. \end{aligned} \tag{34}$$

Hence

$$\operatorname{ran} \mathcal{T} \subseteq \delta_e^\perp, \qquad P_{\delta_e^\perp} = I_{UVQ} - \Pi_e.$$

Since Equation (33) says that $\|\mathcal{T}\|^2 \le (r-1)\beta_2(\Phi)$, for every vector x,

$$\begin{aligned}\langle x, \mathcal{T}\mathcal{T}^* x\rangle &= \langle P_{\delta_e^\perp} x, \mathcal{T}\mathcal{T}^* P_{\delta_e^\perp} x\rangle \\ &\le (r-1)\beta_2(\Phi) \|P_{\delta_e^\perp} x\|^2. \end{aligned}$$

Equivalently,

$$\mathcal{T}\mathcal{T}^* \preceq (r-1)\beta_2(\Phi)(I_{UVQ} - \Pi_e).$$

This is Equation (15), and Equation (12) specializes it to Equation (25), whence $H_r^{(0)} \succeq 0$ by Equation (24). □

Proof of Theorem 2.4. The argument just given uses Φ only through three facts, and they enter at separate points. That Φ is a Kraus sum gives $(\Phi \otimes \mathrm{id}_Q)(P_+) \succeq 0$ and Equation (27). That its Kraus span obeys Equation (32) — which needs only $\beta_2(\Phi)$ — gives Equation (33), hence $\mathcal{T}\mathcal{T}^* \preceq (r-1)\beta_2(\Phi) I$, which is Equation (13). Transpose symmetry of the Kraus operators enters only in Equation (34), and only to replace I by $I - \Pi_e$, giving Equation (15).

Each of Equations (14) and (16) follows from the corresponding upper bound: $\rho_0^{T_{UV}} = P_+ - P_-$ and $(\Phi \otimes \mathrm{id}_Q)(P_+) \succeq 0$, so $(\Phi \otimes \mathrm{id}_Q)(\rho_0^{T_{UV}}) \succeq -(\Phi \otimes \mathrm{id}_Q)(P_-)$.

For the positivity assertion at the size of the displayed frame no further identity is needed. Since the e_i are orthonormal, $\mathrm{Tr}_{UV}\, \rho_0 = I_Q$ and $\rho_0 = r\Pi_e$, so

$$I - \Pi_e = \left[(\Delta - \tfrac{1}{r}\mathrm{id}) \otimes \mathrm{id}_Q\right](\rho_0), \qquad (\Phi \circ \tau \otimes \mathrm{id}_Q)(\rho_0) = (\Phi \otimes \mathrm{id}_Q)\big(\rho_0^{T_{UV}}\big),$$

so Equation (16) is exactly $(\Psi_{r,\Phi} \otimes \mathrm{id}_Q)(\rho_0) \succeq 0$.

It remains to avoid padding a Schmidt decomposition by an orthonormal r-frame, which need not exist when $r > \dim H$. Let a nonzero test vector have exact Schmidt rank $s \le r$, and write it as $(I \otimes D)\delta_e$ for an orthonormal s-frame, using Equation (43). The preceding argument at size s, conjugated by $I \otimes D$, proves nonnegativity for $\Psi_{s,\Phi}$. If C is the matrix obtained by inverse vectorization of the test vector, then the difference of the two Choi quadratic forms is

$$(r-s)\beta_2(\Phi) \left(\|C\|_2^2 - \frac{|\mathrm{Tr}\, C|^2}{rs} \right) \ge 0,$$

because $|\mathrm{Tr}\, C|^2 \le s\|C\|_2^2$. Thus the same vector is nonnegative for $\Psi_{r,\Phi}$. The zero vector is immediate. This proves the assertion for every r, including $r > \dim H$. □

Remark 2.6 (Transpose parity, and where the trace term comes from). Since $\mathrm{vec}(K^T) = F\,\mathrm{vec}(K)$ for the flip F on the two vectorization slots, transposition is an involution of the Kraus space and grades it by the flip eigenvalue: $\wedge^2 H$ has degree 1 and $\mathrm{Sym}^2 H$ degree 0. Kraus spaces multiply under $\otimes$ and the degree is additive, so the grading is by $\mathbb{Z}/2$. In this language:

The hypothesis of Equation (34) is that the Kraus space has degree 0; each $\mathfrak{so}$ factor has degree 1; hence exactly the products of an *even* number of skew factors qualify.

Concretely, every K in the span of $\{L \otimes M\}$ satisfies $K^T = K$, which is what makes the two terms of Equation (34) cancel. In odd degree $K^T = -K$, the two terms add, and $\operatorname{ran}\mathcal{T}$ does not lie in $\delta_e^\perp$; the argument then yields only $\mathcal{T}\mathcal{T}^* \preceq (r-1)I$ in place of $(r-1)(I - \Pi_e)$, giving $rI + \rho_0 - \iota_{UQ}(\rho^0_{UQ}) - \iota_{VQ}(\rho^0_{VQ}) \succeq 0$ and hence the coefficient 1 rather than the sharp $1/r$ in front of $|\operatorname{Tr} C|^2$.

So the $\frac{1}{r}|\operatorname{Tr} C|^2$ term of Theorem 1.2 is exactly the contribution of the single protected direction δ_e, and the grading is what protects it: the generalizations that keep the sharp trace coefficient are the even-degree ones.

Remark 2.7 (The marginal inequality as an r-positivity statement). Recall Δ and τ from Theorem 2.4, and $\mathcal{R}_\alpha(X) = \operatorname{Tr}(X)I + \alpha X$ from Corollary 4.6. Put

$$S := \mathcal{R}_{-1} = \Delta - \mathrm{id}, \qquad t := \frac{r-1}{r}, \qquad \text{so that} \qquad \mathcal{R}_{-1/r} = S + t\,\mathrm{id}.$$

Two facts suffice. First, both Δ and τ factor across tensor products: $\tau_{U\otimes V} = \tau_U \otimes \tau_V$, and $\Delta_{U\otimes V} = \Delta_U \otimes \Delta_V$ because discarding a composite system and repreparing it is discarding and repreparing each factor. Since $\Lambda(Y) = \operatorname{Tr}(Y)I - Y^T$ gives $\Lambda \circ \tau = \Delta - \mathrm{id} = S$ on a single factor,

$$(\Lambda_U \otimes \Lambda_V) \circ \tau = (\Lambda_U \circ \tau_U) \otimes (\Lambda_V \circ \tau_V) = S_U \otimes S_V. \tag{35}$$

Second, $\{\mathrm{id}, \Delta\}$ spans a two-dimensional commutative algebra, and the rest is a binomial expansion in it: using $\Delta = S + \mathrm{id}$ on each factor,

$$\begin{aligned} S \otimes S + (r-1)\,\Delta_U \otimes \Delta_V - t\,\mathrm{id} &= S \otimes S + (r-1)(S + \mathrm{id}) \otimes (S + \mathrm{id}) - t\,\mathrm{id} \\ &= r\,S \otimes S + (r-1)(S \otimes \mathrm{id} + \mathrm{id} \otimes S) + \tfrac{(r-1)^2}{r}\,\mathrm{id} \\ &= r\big[S \otimes S + t(S \otimes \mathrm{id} + \mathrm{id} \otimes S) + t^2\,\mathrm{id}\big], \end{aligned}$$

by $rt = r-1$ and $rt^2 = (r-1)^2/r$. The bracket is $(S + t\,\mathrm{id})^{\otimes 2}$, so for every $r \geq 1$

$$(\Lambda_U \otimes \Lambda_V) \circ \tau + (r-1)\left(\Delta - \tfrac{1}{r}\mathrm{id}\right) = r\,\mathcal{R}^U_{-1/r} \otimes \mathcal{R}^V_{-1/r}. \tag{36}$$

Now $\operatorname{Tr}_{UV} \rho_0 = I_Q$ because the e_i are orthonormal, and $\rho_0 = r\Pi_e$, so $I - \Pi_e = \big[(\Delta - \frac{1}{r}\mathrm{id}) \otimes \mathrm{id}_Q\big](\rho_0)$. Combining this with Equation (23) and Equation (36) rewrites Equation (17) as

$$H_r^{(0)} = \big[(r\,\mathcal{R}^U_{-1/r} \otimes \mathcal{R}^V_{-1/r}) \otimes \mathrm{id}_Q\big](\rho_0).$$

Thus Proposition 2.5 states precisely that $\mathcal{R}_{-1/r} \otimes \mathcal{R}_{-1/r}$ is r-positive. The passage from orthonormal frames $e_1, \ldots, e_r$ to arbitrary vectors of Schmidt rank at most r is the range factorization Equation (43), which produces the frame $\{e_i\}$ and the test vector δ_d in one step; no filtering or limiting argument is involved. Equivalently, this is the sufficiency half of Corollary 4.9 at $a = b = 1/r$, available here, before Theorem 1.2.

Remark 2.8 (Why the correction term has this shape). $\mathcal{L}(H) = \mathbb{C}I \oplus \{\text{traceless}\}$ is a decomposition into inequivalent $\mathrm{U}(d)$-representations, so the commutant — the $\mathrm{U}(d)$-covariant superoperators on $\mathcal{L}(H)$ — is exactly $\operatorname{span}\{\mathrm{id}, \Delta\}$, of dimension two, with $\mathcal{R}_{-a}$ acting as $d - a$ on $\mathbb{C}I$ and as $-a$ on the traceless part. A covariant correction to $\Phi \circ \tau$ therefore lies in a one-parameter pencil once its scale is fixed, and $1/r$ is the single free coefficient; Remark 2.6 identifies what pins it, namely the protected direction δ_e. The correction term is not a choice. The geometry of that pencil — the segment from id to Δ, its extension, and the nested k-positive cones along it — is the subject of [12], and $\mathcal{R}_{-1/r}$ is its r-positivity boundary.

The same count explains the shape of the *conclusion*, not merely of the correction. On a bipartite space the covariant superoperators are the tensor square of the above, of dimension four, and the four basis elements evaluate on C to

$$\|C\|_2^2, \qquad \|\operatorname{Tr}_U C\|_2^2, \qquad \|\operatorname{Tr}_V C\|_2^2, \qquad |\operatorname{Tr} C|^2.$$

So every Hermitian sesquilinear functional of degree $(1,1)$ in $(C,\overline{C})$ that is invariant under local unitaries $C \mapsto (U\otimes V)C(U\otimes V)^*$ is a combination of exactly these four, and Theorem 1.2 is an inequality between the only quantities available within that class. The qualifier is essential: real quadratic expressions such as $\operatorname{Re}\operatorname{Tr}(C^2)$ obey different representation-theoretic rules and are not classified by this commutant count. This is also why the same four terms recur in otherwise unrelated settings — Gaussian fourth moments with product structure, subsystem purities — since a Haar average of a Hermitian sesquilinear form lands in the commutant by construction. The hypothesis is the full local unitary group: a quantity invariant only under the stabilizer of some fixed structure has a larger commutant and need not lie in this span.

The collapse in Equation (36) is specific to two factors. For n factors the same factorization gives $\Delta = \sum_{T\subseteq[n]} \bigotimes_{i\in T} S_i$, and matching against $c\bigotimes(S+t\,\mathrm{id})$ forces $r-1 = r\,t^{n-|T|}$ for every T with $|T|<n$. At $|T|=n-1$ this reads $r-1=rt$, which holds; for $n=2$ the $-t\,\mathrm{id}$ term repairs $|T|=0$; but for $n\geq 3$ the equation at $|T|=n-2$ is $r-1=(r-1)^2/r$, i.e. $r=1$. So the even-partite generalizations of Remark 2.6 are genuine inequalities and cannot be recast as a map identity of the form $\bigotimes \mathcal{R}_{-a}$.

3. Proof of the Rank-r Partial-Trace Inequality

Lemma 3.1 (Partial-trace contraction identities)**.** *Let $e_1,\dots,e_s \in U\otimes V$ be orthonormal, let $d_1,\dots,d_s\in U\otimes V$, and let $q_1,\dots,q_s$ be an orthonormal basis of $Q=\mathbb{C}^s$. Define*

$$C := \sum_{i=1}^s |e_i\rangle\langle d_i|, \qquad \delta_e := \sum_{i=1}^s e_i\otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle\delta_e|,$$

and

$$\delta_d := \sum_{i=1}^s d_i\otimes q_i, \qquad \rho^0_{UQ} := \operatorname{Tr}_V \rho_0, \qquad \rho^0_{VQ} := \operatorname{Tr}_U \rho_0.$$

Then

$$\langle\delta_d,\delta_d\rangle = \|C\|_2^2, \tag{37}$$
$$\langle\delta_d,\rho_0\delta_d\rangle = |\operatorname{Tr} C|^2, \tag{38}$$
$$\langle\delta_d,\iota_{UQ}(\rho^0_{UQ})\delta_d\rangle = \|\operatorname{Tr}_U C\|_2^2, \tag{39}$$
$$\langle\delta_d,\iota_{VQ}(\rho^0_{VQ})\delta_d\rangle = \|\operatorname{Tr}_V C\|_2^2. \tag{40}$$

Proof. The first identity follows from the orthonormality of the e_i:

$$\|C\|_2^2 = \sum_i \|d_i\|^2 = \langle\delta_d,\delta_d\rangle.$$

Since $\operatorname{Tr} C = \sum_i \langle d_i, e_i\rangle$ under the stated inner-product convention,

$$\langle\delta_e,\delta_d\rangle = \sum_i \langle e_i, d_i\rangle = \overline{\operatorname{Tr} C},$$

which proves Equation (38).

For the first marginal contraction,

$$\rho^0_{UQ} = \sum_{i,j} \operatorname{Tr}_V |e_i\rangle\langle e_j| \otimes |q_i\rangle\langle q_j|,$$

and hence

$$\langle \delta_d, \iota_{UQ}(\rho^0_{UQ})\delta_d\rangle = \sum_{i,j} \langle d_i, (\mathrm{Tr}_V |e_i\rangle\langle e_j| \otimes I_V) d_j\rangle. \tag{41}$$

Choose product bases $\{u_a\}$ of U and $\{v_b\}$ of V, and write

$$e_i = \sum_{a,b} e_{i,ab} u_a \otimes v_b, \qquad d_i = \sum_{a,b} d_{i,ab} u_a \otimes v_b.$$

Then

$$(\mathrm{Tr}_U C)_{b,c} = \sum_{i,a} e_{i,ab}\overline{d_{i,ac}},$$

so

$$\|\mathrm{Tr}_U C\|_2^2 = \sum_{i,j}\sum_{a,a'}\sum_{b,c} e_{i,ab}\overline{d_{i,ac}}\,\overline{e_{j,a'b}} d_{j,a'c}. \tag{42}$$

Expanding the right-hand side of Equation (41) gives, explicitly,

$$\begin{aligned}\sum_{i,j} &\langle d_i, (\mathrm{Tr}_V |e_i\rangle\langle e_j| \otimes I_V) d_j\rangle \\ &= \sum_{i,j}\sum_{a,a'}\sum_{b,c} e_{i,ab}\overline{d_{i,ac}}\,\overline{e_{j,a'b}} d_{j,a'c},\end{aligned}$$

the same quadruple sum as in Equation (42). This proves Equation (39). Interchanging U and V proves Equation (40). □

Proof of Theorem 1.2. The case $C = 0$ is immediate. Suppose $C \neq 0$, and let

$$r_0 := \operatorname{rank} C.$$

Choose an orthonormal basis $\{e_i\}_{i=1}^{r_0}$ of $\operatorname{ran} C$ and set

$$d_i := C^* e_i.$$

Then

$$C = \sum_{i=1}^{r_0} |e_i\rangle\langle d_i|, \qquad \sum_{i=1}^{r_0} \|d_i\|^2 = \|C\|_2^2. \tag{43}$$

Since $\operatorname{rank} C \leq r$, one has $1 \leq r_0 \leq r$.

Let $Q = \mathbb{C}^{r_0}$, with orthonormal basis $\{q_i\}_{i=1}^{r_0}$, and define

$$\delta_e := \sum_{i=1}^{r_0} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle\delta_e|, \qquad \Pi_e := \frac{1}{r_0}\rho_0, \qquad \delta_d := \sum_{i=1}^{r_0} d_i \otimes q_i.$$

Write $\rho^0_{UQ} = \mathrm{Tr}_V \rho_0$ and $\rho^0_{VQ} = \mathrm{Tr}_U \rho_0$. Applying Proposition 2.5 with $r = r_0$ gives $H^{(0)}_{r_0} \succeq 0$. Therefore,

$$\begin{aligned} 0 &\leq \langle \delta_d, H^{(0)}_{r_0} \delta_d\rangle \\ &= r_0\langle \delta_d, \delta_d\rangle + \frac{1}{r_0}\langle \delta_d, \rho_0 \delta_d\rangle - \langle \delta_d, \iota_{UQ}(\rho^0_{UQ})\delta_d\rangle - \langle \delta_d, \iota_{VQ}(\rho^0_{VQ})\delta_d\rangle. \end{aligned} \tag{44}$$

By Lemma 3.1, this is precisely

$$\|\mathrm{Tr}_U C\|_2^2 + \|\mathrm{Tr}_V C\|_2^2 \leq r_0\|C\|_2^2 + \frac{1}{r_0}|\mathrm{Tr}\, C|^2. \tag{45}$$

It remains to pass from r_0 to r. From Equation (43) and $\operatorname{Tr} C = \sum_i \langle d_i, e_i\rangle$,

$$|\operatorname{Tr} C|^2 \le \left(\sum_{i=1}^{r_0} \|d_i\|\right)^2 \le r_0 \sum_{i=1}^{r_0} \|d_i\|^2 = r_0 \|C\|_2^2.$$

The passage from r_0 to r is then a single identity: for all $r \ge r_0 \ge 1$,

$$r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \left(r_0\|C\|_2^2 + \frac{1}{r_0}|\operatorname{Tr} C|^2\right) = (r - r_0)\left(\|C\|_2^2 - \frac{|\operatorname{Tr} C|^2}{r r_0}\right). \tag{46}$$

By the trace–rank bound the second factor is at least $\left(1 - \frac{1}{r}\right)\|C\|_2^2 \ge 0$, and the first is nonnegative, so the right-hand side is nonnegative. Combining this with Equation (45) proves Equation (3).

Both factors are moreover *strictly* positive when $C \ne 0$ and $r_0 < r$: the first is at least 1, and $r_0 < r$ forces $r \ge 2$, so $1 - \frac{1}{r} \ge \frac{1}{2}$ while $\|C\|_2^2 > 0$. Hence the inequality of Equation (3) is strict whenever $\operatorname{rank} C < r$ and $C \ne 0$, which is the exact-rank assertion of Remark 1.3. ☐

Remark 3.2 (What β_2 does and does not carry). Theorem 2.4 depends on Φ only through $\beta_2(\Phi)$. The passage from it to a *partial-trace* inequality does not: the marginals $\operatorname{Tr}_U C$ and $\operatorname{Tr}_V C$ appear only through Equation (23), a contraction identity for the particular map $\Lambda_U \otimes \Lambda_V$, and no such identity is available for a general Φ. A second map with the same β_2 contracts against δ_d to a different quadratic form.

That β_2 alone cannot suffice is visible at $\Phi = 0$, which satisfies every hypothesis of Theorem 2.4 with $\beta_2 = 0$: the extremizer $C = P_r \otimes |v\rangle\langle w|$ of Remark 1.3 with $v \perp w$ has $\|\operatorname{Tr}_U C\|_2^2 = r^2$, $\operatorname{Tr}_V C = 0$, $\|C\|_2^2 = r$ and $\operatorname{Tr} C = 0$, so no inequality with Φ-independent coefficients degenerating at $\beta_2 = 0$ can hold. What survives at general β_2 is the operator statement Equations (15) and (16) and, for $\Phi = \Lambda_U \otimes \Lambda_V$ with the constant left free, the family

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le \left[r + (\beta_2 - 1)(r-1)\right]\|C\|_2^2 + \left[1 - (\beta_2 - 1)(r-1)\right]\frac{1}{r}|\operatorname{Tr} C|^2 \tag{47}$$

for nonzero C with $r = \operatorname{rank} C$, obtained by carrying β_2 rather than 1 through Equation (33). At $\beta_2 = 1$ this is Equation (2).

The coefficient $(r-1)\beta_2$ in Theorem 2.4 bounds the correction rather than computing it: the vertex step Equation (30) discards the nonnegative variance terms of Proposition B.2, so for a general Φ a smaller multiple of $\Delta - \frac{1}{r}\mathrm{id}$ may already suffice. For $\Phi = \Lambda_U \otimes \Lambda_V$ the value $\beta_2 = 1$ cannot be improved: as noted after Equation (33) it is already attained at $\dim U = \dim V = 2$, where $\mathfrak{so}(2) \otimes \mathfrak{so}(2)$ is one-dimensional and every orthonormal pair saturates Equation (4). The extremizer of Remark 1.3 forces $\beta_2 \ge 1$ there unconditionally, so Equation (47) is never stronger than Theorem 1.2 on that space.

Remark 3.3 (Instances). An instance of Theorem 1.1 is a completely positive Φ admitting a transpose-symmetric Kraus representation, together with the value of $\beta_2(\Phi)$; a partial-trace consequence needs in addition a contraction identity of the kind Equation (23) supplies for $\Lambda_U \otimes \Lambda_V$. The two directions one would look in next each stop at a definite point, and it is worth recording where.

Schmidt rank p. Equation (31) generalizes at once: a rank-p compression P and the $S(p)$-norm of $J(\Phi)$ give $\sum_{i \le p} \|K x_i\|^2 \le p\, \beta_p(\Phi) \|c\|_2^2$ for any orthonormal p-tuple. The reduction does not. The pair structure of Section 3 comes from the partial transpose and not from the Schmidt rank: $P_\pm$ are the ± 1 eigenspaces of the flip that $\rho_0^{T_{UV}}$ decomposes into, and η_{ij}, ζ_{ij}, and the edge set of K_r are pair-indexed downstream of that. A p-fold analogue would have to decompose the symmetric group S_p, which is a different problem. We record it as open.

Higher copies. By Remark 2.6, $\Phi = \Lambda_{U_1} \otimes \cdots \otimes \Lambda_{U_{2m}}$ has degree 0 and so satisfies the hypothesis of Theorem 2.4 for every m; expanding $\Phi \circ \tau$ then gives, for $\operatorname{rank} C \le r$,

$$\sum_{S \subseteq [2m]} (-1)^{|S|} \|\mathrm{Tr}_S\, C\|_2^2 + (r-1)\beta_2(\Phi)\left(\|C\|_2^2 - \tfrac{1}{r}|\mathrm{Tr}\, C|^2\right) \ge 0.$$

At $m = 1$ this is Equation (2). What is missing is the constant: $\beta_2(\Lambda^{\otimes 2m})$ for $m \ge 2$ is a Ky-Fan-2 concentration constant for the $2m$-fold antisymmetric projection, and we do not know its value. So the family *reduces* to one rank-two Choi $S(2)$-norm per m; it does not follow from Lemma 2.1. By Remark 2.8 it also cannot be recast as a map identity.

4. Applications and Corollaries

This section collects consequences of the results proved above: the exact symmetric score curve, an asymmetric tensor-product score theorem, block positivity and positive maps, quantitative Schmidt-number separation, Ky Fan bounds for Kronecker sums, and the exact Schmidt number of a product of antisymmetric projector states. All of them descend from Theorem 1.2, or from Lemma 2.1, rather than from Theorem 1.1 directly. The development checks the two score curves as pointwise lower bounds plus rank-exact attaining matrices, their rank-constrained nonnegativity thresholds, the adjacent-rank signed separation, the exact coordinate Choi matrices of the reduction maps and their regrouped tensor products, block-positivity thresholds at every positive rank in equal local dimensions, the equivalence with positivity of the actual finite-coordinate map ampliations, the resulting symmetric and asymmetric map classifications, and the rank-sensitive estimate Equation (67). The trace-norm and Ky Fan dualities and the mixed-state Schmidt-number arguments are not formalized.

The consequences at $r = 2$ — the two-copy distillability threshold and the statements at the endpoint $\alpha = -\frac{1}{2}$ — are also reached in [9, 8, 10]. Song and Chen include a two-parameter rank-two extension, and Bharti, Gajjala and Haug give finite-copy extensions beyond anything proved here. Relative to those contemporaneous results, the additional content developed below is the exact score curve at every r, the all-r asymmetric classification, and the Schmidt-number consequences that require $r > 2$.

They do not all rest on the same input. Proposition 4.1, Theorem 4.8, and Corollary 4.11 use Theorem 1.2 at general r, and are the results for which the extension beyond normal matrices is what is needed; Corollaries 4.5, 4.6 and 4.9 restate them through the Choi–Jamiołkowski correspondence. Corollaries 4.12 and 4.13, by contrast, rest on Lemma 2.1 alone and do not use Theorem 1.2. Appendix C collects two further consequences that use nothing about the witnesses constructed here beyond r-block positivity.

4.1. **Two-copy block positivity.**

Fix $d \ge 2$ and $\alpha \in \mathbb{R}$. Let

$$\rho_\alpha := I + \alpha F$$

be the unnormalized Werner operator on $\mathbb{C}^d \otimes \mathbb{C}^d$; for $\alpha \in [-1, 1]$, it is proportional to the standard Werner state [16]. Its partial transpose is

$$\rho_\alpha^\Gamma = I + \alpha|\Omega\rangle\langle\Omega|, \qquad \Omega := \sum_{i=1}^d e_i \otimes e_i.$$

The case $\alpha \ge 0$ is immediate because

$$I + \alpha|\Omega\rangle\langle\Omega| \succeq 0.$$

For $\alpha \le 0$ and $C \in M_{d^2}(\mathbb{C})$, define

$$q^{(2)}(\alpha, C) := \|C\|_2^2 + \alpha\left(\|\mathrm{Tr}_1\, C\|_2^2 + \|\mathrm{Tr}_2\, C\|_2^2\right) + \alpha^2|\mathrm{Tr}\, C|^2. \tag{48}$$

The first argument is the Werner parameter of ρ_α; only $\alpha \le 0$ is at issue, and it is convenient to write $\alpha = -t$ with $t \ge 0$, so that

$$q^{(2)}(-t, C) = \|C\|_2^2 - t\left(\|\mathrm{Tr}_1 C\|_2^2 + \|\mathrm{Tr}_2 C\|_2^2\right) + t^2 |\mathrm{Tr}\, C|^2.$$

An operator is called r-block-positive if its expectation is nonnegative on every vector of Schmidt rank at most r. With the vectorization convention fixed in the introduction and the regrouping $(A_1A_2) : (B_1B_2)$, one has

$$\begin{aligned} &\left\langle \mathrm{vec}(C), (\rho_\alpha^\Gamma)^{\otimes 2}\, \mathrm{vec}(C) \right\rangle \\ &= q^{(2)}(\alpha, C), \end{aligned} \tag{49}$$

which is what the first argument of $q^{(2)}$ records. Moreover,

$$\mathrm{SR}(\mathrm{vec}(C)) = \mathrm{rank}\, C$$

across the same output–input regrouping. Thus r-block positivity of $(\rho_{-t}^\Gamma)^{\otimes 2}$ is equivalent to $q^{(2)}(-t, C) \ge 0$ for every matrix C of rank at most r. For every C of rank at most r, Equation (3) implies

$$q^{(2)}(-t, C) \ge (1 - rt)\|C\|_2^2 + \left(t^2 - \frac{t}{r}\right) |\mathrm{Tr}\, C|^2. \tag{50}$$

If $0 \le t \le 1/r$, then $t^2 - t/r \le 0$. Using

$$|\mathrm{Tr}\, C|^2 \le r\|C\|_2^2$$

in the appropriate direction gives

$$\begin{aligned} q^{(2)}(-t, C) &\ge \left(1 - rt + rt^2 - t\right) \|C\|_2^2 \\ &= (1 - t)(1 - rt)\|C\|_2^2 \ge 0. \end{aligned} \tag{51}$$

Proposition 4.1 (Exact rank-constrained two-copy score curve)**.** *Let $d \ge 2$ and $1 \le r \le d$. For $t \ge 0$, define*

$$\mu_r(t) := \inf_{\substack{C \in M_{d^2}(\mathbb{C}) \setminus \{0\} \\ \mathrm{rank}\, C \le r}} \frac{q^{(2)}(-t, C)}{\|C\|_2^2}.$$

Then

$$\mu_r(t) = \begin{cases} (1 - t)(1 - rt), & 0 \le t \le 1/r, \\ 1 - rt, & t \ge 1/r. \end{cases} \tag{52}$$

Both branches are attained by rank-r matrices.

Proof. For $C \ne 0$ of rank at most r, set

$$z := \frac{|\mathrm{Tr}\, C|^2}{\|C\|_2^2}.$$

The trace–rank estimate gives $0 \le z \le r$. Dividing Equation (50) by $\|C\|_2^2$ yields

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \ge 1 - rt + \left(t^2 - \frac{t}{r}\right) z. \tag{53}$$

If $0 \le t \le 1/r$, the coefficient of z is nonpositive, so Equation (53) and $z \le r$ give

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \ge 1 - rt + r\left(t^2 - \frac{t}{r}\right) = (1 - t)(1 - rt).$$

Choose a rank-r projection $P_r \in M_d(\mathbb{C})$ and a unit vector $v \in \mathbb{C}^d$. For

$$C = P_r \otimes |v\rangle\langle v|$$

one has

$$\|C\|_2^2 = r, \qquad |\operatorname{Tr} C|^2 = r^2,$$
$$\|\operatorname{Tr}_1 C\|_2^2 = r^2, \quad \|\operatorname{Tr}_2 C\|_2^2 = r,$$

and hence equality holds.

If $t \geq 1/r$, the coefficient of z in Equation (53) is nonnegative, so

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \geq 1 - rt.$$

Choose orthogonal unit vectors $v, w \in \mathbb{C}^d$ and set

$$C = P_r \otimes |v\rangle\langle w|.$$

Then

$$\operatorname{Tr} C = 0, \qquad \|\operatorname{Tr}_1 C\|_2^2 = r^2, \qquad \operatorname{Tr}_2 C = 0,$$

so equality again holds. $\square$

Remark 4.2 (Why the range $r \leq d$ is essential). The extremizers in Proposition 4.1 require a rank-r projection in one d-dimensional tensor factor. The score formula does not in general extend to $r > d$ by replacing r with $\min\{r, d\}$. For example, when the rank bound is $r = d^2$, no rank constraint remains. If $0 \leq t \leq 1/d$, the smallest eigenvalue of

$$\left(\rho_{-t}^{\Gamma}\right)^{\otimes 2}$$

is

$$(1 - dt)^2.$$

By Equation (49), the displayed infimum is the smallest eigenvalue, and therefore

$$\inf_{C \neq 0} \frac{q^{(2)}(-t, C)}{\|C\|_2^2} = (1 - dt)^2.$$

This generally differs from $(1-t)(1-dt)$, the expression obtained by formally substituting d for r in the first branch of Equation (52).

Corollary 4.3 (An explicit adjacent-Schmidt-rank violation)**.** *Let $1 \leq r < d$ and put*

$$Z_{r,d} := \left(I - \frac{1}{r}|\Omega\rangle\langle\Omega|\right)^{\otimes 2}.$$

Across the regrouped bipartition $(A_1A_2) : (B_1B_2)$,

$$\langle \psi, Z_{r,d}\psi\rangle \geq 0 \qquad \text{whenever } \|\psi\| = 1 \text{ and } \operatorname{SR}(\psi) \leq r.$$

On the other hand, there is a unit vector ψ_{r+1} of Schmidt rank $r+1$ such that

$$\langle \psi_{r+1}, Z_{r,d}\psi_{r+1}\rangle = -\frac{1}{r}. \tag{54}$$

Proof. The nonnegativity assertion is Corollary 4.5 at $t = 1/r$. Choose a rank-$(r+1)$ projection P_{r+1} and orthogonal unit vectors $v, w \in \mathbb{C}^d$, and define

$$C = P_{r+1} \otimes |v\rangle\langle w|, \qquad \psi_{r+1} := \frac{\operatorname{vec}(C)}{\sqrt{r+1}}.$$

Then $\operatorname{SR}(\psi_{r+1}) = \operatorname{rank} C = r + 1$, while

$$\frac{q^{(2)}(-1/r, C)}{\|C\|_2^2} = 1 - \frac{r+1}{r} = -\frac{1}{r}.$$

Now use Equation (49). $\square$

Remark 4.4 (A strict signed separation). Let $0 < \varepsilon < 1/(r+1)$ and set $t = (1-\varepsilon)/r$. Then Proposition 4.1 gives

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \geq \varepsilon \left(1 - \frac{1-\varepsilon}{r}\right) > 0 \qquad (\operatorname{rank} C \leq r),$$

whereas the rank-$(r+1)$ matrix used in Corollary 4.3 has normalized score

$$1 - (r+1)t = \frac{(r+1)\varepsilon - 1}{r} < 0.$$

Thus the filter has a uniform positive margin on all normalized matrices of rank at most r, while the displayed rank-$(r+1)$ matrix has negative score.

This yields the exact r-block-positivity threshold for the two-copy family $(\rho_\alpha^\Gamma)^{\otimes 2}$.

Corollary 4.5 (Two-copy block positivity)**.** *Let $d \geq 2$ and $\alpha \in \mathbb{R}$. For every $1 \leq r \leq d^2$, the operator*

$$(\rho_\alpha^\Gamma)^{\otimes 2}$$

is r-block-positive across the regrouped bipartition

$$(A_1 A_2) : (B_1 B_2)$$

if and only if

$$\alpha \geq -\frac{1}{\min\{r, d\}}. \tag{55}$$

Proof. It remains only to consider $\alpha = -t < 0$.

Suppose first that $r \leq d$. If $t \leq 1/r$, then Equation (51) gives

$$q^{(2)}(-t, C) \geq 0$$

for every C of rank at most r, proving r-block positivity.

For necessity in all cases, set

$$s := \min\{r, d\}.$$

Choose a rank-s projection P_s and orthogonal unit vectors $v, w \in \mathbb{C}^d$, and let

$$C = P_s \otimes |v\rangle\langle w|.$$

Then

$$\begin{aligned} \operatorname{rank} C &= s \leq r, \\ \operatorname{Tr} C &= 0, \\ \|C\|_2^2 &= s, \\ \|\operatorname{Tr}_1 C\|_2^2 &= s^2, \\ \|\operatorname{Tr}_2 C\|_2^2 &= 0. \end{aligned}$$

Hence

$$q^{(2)}(-t, C) = s(1 - st),$$

which is negative whenever $t > 1/s$. Thus r-block positivity forces

$$t \leq \frac{1}{\min\{r, d\}}.$$

Finally, suppose $r > d$. If $t \leq 1/d$, then the Choi operator of

$$\mathcal{R}_{-t}(X) := \operatorname{Tr}(X) I - tX$$

is

$$J(\mathcal{R}_{-t}) = I - t|\Omega\rangle\langle\Omega| \succeq 0,$$

because $\|\Omega\|^2 = d$ and the only potentially nonpositive eigenvalue is $1 - td$. Therefore $\mathcal{R}_{-t}$ is completely positive, so $\mathcal{R}_{-t}^{\otimes 2}$ is completely positive and hence r-positive for every r. This proves sufficiency when $r > d$. □

4.2. **r-positivity of the tensor square.** The preceding block-positivity statement has an equivalent map formulation through the Choi–Jamiołkowski correspondence: a map is r-positive if and only if its Choi operator is r-block-positive [1, 15]. With the output factors grouped against the input factors, the Choi operator of $\mathcal{R}_\alpha^{\otimes 2}$ is exactly $(\rho_\alpha^\Gamma)^{\otimes 2}$ across the bipartition $(A_1A_2) : (B_1B_2)$ used above.

Corollary 4.6 (Tensor-square stability)**.** *Let $d \geq 2$ and $\alpha \in \mathbb{R}$. Define*

$$\mathcal{R}_\alpha(X) := \operatorname{Tr}(X)I + \alpha X.$$

For every $1 \leq r \leq d^2$, the tensor-square map

$$\mathcal{R}_\alpha^{\otimes 2}$$

is r-positive if and only if

$$\alpha \geq -\frac{1}{\min\{r, d\}}.$$

The single factor is classical: $\mathcal{R}_\alpha$ on M_d is k-positive exactly when $\alpha \geq -1/k$, which is [12, Thm. 2(i)] after reparametrization, and $\mathcal{R}_{-1/k}$ is the boundary map he denotes τ_k. The tensor square does not follow from it, positivity being unstable under tensoring; the maps whose positivity survives all of their own tensor powers are studied in [14]. For $r \geq d$ the statement below is immediate, each factor being then completely positive, so the content is the range $r < d$.

4.3. **Exact asymmetric tensor-product theorem.** For $a, b \geq 0$, define

$$q_{a,b}(C) := \|C\|_2^2 - a\|\operatorname{Tr}_1 C\|_2^2 - b\|\operatorname{Tr}_2 C\|_2^2 + ab|\operatorname{Tr} C|^2. \tag{56}$$

Under the same vectorization and regrouping as above,

$$q_{a,b}(C) = \langle \operatorname{vec}(C), (I - a|\Omega\rangle\langle\Omega|) \otimes (I - b|\Omega\rangle\langle\Omega|) \operatorname{vec}(C)\rangle .$$

Lemma 4.7 (One-sided rank bound)**.** *If* $\operatorname{rank} C \leq r$*, then*

$$\|\operatorname{Tr}_j C\|_2^2 \leq r\|C\|_2^2, \qquad j \in \{1, 2\}. \tag{57}$$

Proof. Write $C = \sum_{i=1}^{r_0} |e_i\rangle\langle d_i|$ as in Equation (43), with $e_1, \ldots, e_{r_0}$ orthonormal and $r_0 = \operatorname{rank} C \leq r$. Identify a vector x of the bipartite space with its coefficient matrix X relative to the chosen product basis, so that $\|X\|_2 = \|x\|$. The two partial traces of a rank-one operator are then matrix products,

$$\operatorname{Tr}_2|x\rangle\langle y| = XY^*, \qquad \operatorname{Tr}_1|x\rangle\langle y| = \overline{X^*Y}, \tag{58}$$

both immediate from the definitions. The Hilbert–Schmidt norm is submultiplicative, so each summand obeys $\|\operatorname{Tr}_j|e_i\rangle\langle d_i|\|_2 \leq \|e_i\|\|d_i\| = \|d_i\|$. By the triangle inequality and then Cauchy–Schwarz,

$$\|\operatorname{Tr}_j C\|_2^2 \leq \left(\sum_{i=1}^{r_0}\|d_i\|\right)^2 \leq r_0\sum_{i=1}^{r_0}\|d_i\|^2 = r_0\|C\|_2^2 \leq r\|C\|_2^2,$$

the equality being Equation (43). □

Recall that Tr_1 means the trace *over* the first local factor, leaving an operator on the second, while Tr_2 traces over the second and leaves an operator on the first. Thus the coefficient a in Equation (56) multiplies the first-factor trace and b multiplies the second-factor trace.

Theorem 4.8 (Exact asymmetric tensor-product score)**.** *Let $d \geq 2$, $1 \leq r \leq d$, and $a, b \geq 0$. Put*

$$m := \max\{a, b\}, \qquad \ell := \min\{a, b\}.$$

Then

$$\inf_{\substack{C \in M_{d^2}(\mathbb{C}) \setminus \{0\} \\ \operatorname{rank} C \leq r}} \frac{q_{a,b}(C)}{\|C\|_2^2} = \begin{cases} (1-\ell)(1-rm), & m \leq 1/r, \\ 1 - rm, & m \geq 1/r. \end{cases} \tag{59}$$

Proof. By interchanging the two tensor factors, assume $a = m$ and $b = \ell$. For $C \neq 0$ of rank at most r, write

$$x := \frac{\|\mathrm{Tr}_1 C\|_2^2}{\|C\|_2^2}, \qquad y := \frac{\|\mathrm{Tr}_2 C\|_2^2}{\|C\|_2^2}, \qquad z := \frac{|\mathrm{Tr}\, C|^2}{\|C\|_2^2}.$$

By Theorem 1.2 and Lemma 4.7,

$$x + y \leq r + \frac{z}{r}, \qquad x \leq r, \qquad 0 \leq z \leq r.$$

Consequently,

$$\begin{aligned} ax + by &= b(x+y) + (a-b)x \\ &\leq b\left(r + \frac{z}{r}\right) + (a-b)r = ar + \frac{b}{r}z, \end{aligned}$$

and hence

$$\frac{q_{a,b}(C)}{\|C\|_2^2} \geq 1 - ar + b\left(a - \frac{1}{r}\right) z. \tag{60}$$

If $a \leq 1/r$, minimize the right-hand side over $0 \leq z \leq r$ by taking $z = r$; this gives

$$1 - ar + br\left(a - \frac{1}{r}\right) = (1-b)(1-ar).$$

Equality is attained by $C = P_r \otimes |v\rangle\langle v|$. If $a \geq 1/r$, minimize by taking $z = 0$; this gives $1 - ar$, with equality attained by $C = P_r \otimes |v\rangle\langle w|$ for $v \perp w$. If $b > a$, interchange the two tensor factors in these constructions. ∎

Corollary 4.9 (Asymmetric tensor-product r-positivity threshold)**.** *Let $d \geq 2$, $1 \leq r \leq d^2$, and $a, b \geq 0$. Then*

$$(I - a|\Omega\rangle\langle\Omega|) \otimes (I - b|\Omega\rangle\langle\Omega|)$$

is r-block-positive across the regrouped bipartition if and only if

$$\max\{a, b\} \leq \frac{1}{\min\{r, d\}}. \tag{61}$$

Equivalently,

$$\mathcal{R}_{-a} \otimes \mathcal{R}_{-b}$$

is r-positive if and only if Equation (61) holds.

Proof. The corner $a = b = 1/r$ of the sufficiency half is Remark 2.7, which derives it from Proposition 2.5 directly; it does not depend on this section. For the general case with $r \leq d$, the assertion follows from Theorem 4.8. (That route passes through Theorem 1.2, hence through Proposition 2.5; the remark shows the $a = b = 1/r$ corner without the detour. The other corners do need an argument rather than a citation: $\mathcal{R}_{-a} = \mathcal{R}_{-1/r} + (\frac{1}{r} - a)\mathrm{id}$, and the cross terms $\mathcal{R}_{-1/r} \otimes \mathrm{id}$ are not separately positive, so monotonicity in a and b is not immediate.) If $r > d$ and $a, b \leq 1/d$, both factors $I - a|\Omega\rangle\langle\Omega|$ and $I - b|\Omega\rangle\langle\Omega|$ are positive semidefinite, so their tensor product is positive semidefinite. Conversely, if, say, $a > 1/d$, choose

$$C = I_d \otimes |v\rangle\langle w|, \qquad v \perp w.$$

Then $\operatorname{rank} C = d \le r$ and $q_{a,b}(C)/\|C\|_2^2 = 1 - ad < 0$. The case $b > 1/d$ follows by interchanging the tensor factors. The map statement follows from the Choi–Jamiołkowski correspondence. $\square$

4.4. **Quantitative trace-norm separation from bounded Schmidt number.**

Corollary 4.10 (Trace-norm separation certified by the two-copy witness)**.** *Let $1 \le r < d$ and let $Z_{r,d}$ be as in Corollary 4.3. Put*

$$\gamma := \frac{d}{r} - 1 > 0, \qquad \Delta_{r,d} := \gamma + \max\{1, \gamma^2\}.$$

Define the normalized Schmidt-number-r state set

$$\mathcal{S}_r := \{\sigma \succeq 0 : \operatorname{Tr}\sigma = 1,\ \operatorname{SN}(\sigma) \le r\}.$$

If ρ is a state satisfying

$$\operatorname{Tr}(Z_{r,d}\rho) = -\varepsilon < 0,$$

then

$$\inf_{\sigma \in \mathcal{S}_r} \|\rho - \sigma\|_1 \ge \frac{2\varepsilon}{\Delta_{r,d}}. \tag{62}$$

Proof. Since $Z_{r,d}$ is r-block-positive,

$$\operatorname{Tr}(Z_{r,d}\sigma) \ge 0 \qquad \text{for every } \sigma \in \mathcal{S}_r.$$

The operator $I - r^{-1}|\Omega\rangle\langle\Omega|$ has eigenvalues 1 and $1 - d/r = -\gamma$. Hence $Z_{r,d}$ has eigenvalues

$$1, \qquad -\gamma, \qquad \gamma^2.$$

Thus its spectral diameter is $\Delta_{r,d}$.

Fix $\sigma \in \mathcal{S}_r$. Then

$$\operatorname{Tr}\big[Z_{r,d}(\sigma - \rho)\big] \ge \varepsilon.$$

Because $\operatorname{Tr}(\sigma - \rho) = 0$, we may subtract from $Z_{r,d}$ the midpoint of its extreme eigenvalues. The resulting centered operator has operator norm $\Delta_{r,d}/2$. Hölder duality therefore gives

$$\varepsilon \le \frac{\Delta_{r,d}}{2}\|\sigma - \rho\|_1.$$

Taking the infimum over $\sigma \in \mathcal{S}_r$ proves the claim. $\square$

4.5. **Kronecker-sum Ky Fan bounds.** By dualizing the partial-trace inequality over rank-k test operators, Theorem 1.2 yields a singular-value hierarchy for Kronecker sums.

Corollary 4.11. *Let $A, B \in M_d(\mathbb{C})$ be traceless, not necessarily Hermitian, operators, and define*

$$M := A \otimes I_d + I_d \otimes B.$$

For every $1 \le k \le d^2$,

$$\sum_{j=1}^{k} s_j(M)^2 \le \min\left\{d, k + \max\left\{0, 1 - \frac{2k}{d}\right\}\right\}\left(\|A\|_2^2 + \|B\|_2^2\right). \tag{63}$$

In particular, if $d \le 2k$, then

$$\sum_{j=1}^{k} s_j(M)^2 \le k\left(\|A\|_2^2 + \|B\|_2^2\right).$$

Proof. The Frobenius-dual form of Ky Fan duality gives

$$\left(\sum_{j=1}^{k} s_j(M)^2\right)^{1/2} = \max_{\substack{\operatorname{rank} C \le k \\ \|C\|_2 = 1}} |\langle C, M\rangle_{\mathrm{HS}}|. \tag{64}$$

Since A and B are traceless,

$$\begin{aligned}\langle C, M\rangle_{\mathrm{HS}} = &\left\langle \operatorname{Tr}_2 C - \frac{\operatorname{Tr} C}{d} I, A\right\rangle_{\mathrm{HS}} \\ &+ \left\langle \operatorname{Tr}_1 C - \frac{\operatorname{Tr} C}{d} I, B\right\rangle_{\mathrm{HS}}.\end{aligned} \tag{65}$$

By Cauchy–Schwarz,

$$\begin{aligned}|\langle C, M\rangle_{\mathrm{HS}}|^2 \le &\left(\|A\|_2^2 + \|B\|_2^2\right) \\ &\times \left(\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2 - \frac{2}{d}|\operatorname{Tr} C|^2\right).\end{aligned} \tag{66}$$

For a rank-at-most-k operator C with $\|C\|_2 = 1$, Equation (3) gives

$$\begin{aligned}&\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2 - \frac{2}{d}|\operatorname{Tr} C|^2 \\ &\le k + \left(\frac{1}{k} - \frac{2}{d}\right)|\operatorname{Tr} C|^2.\end{aligned} \tag{67}$$

If $1/k - 2/d \le 0$, the final term may be discarded. If $1/k - 2/d > 0$, then

$$|\operatorname{Tr} C|^2 \le k\|C\|_2^2 = k.$$

Thus the right-hand side of Equation (67) is at most

$$k + \max\left\{0, 1 - \frac{2k}{d}\right\}.$$

Combining this with Equations (64) and (66) gives the second entry in the minimum in Equation (63). On the other hand, tracelessness implies

$$\|M\|_2^2 = d\left(\|A\|_2^2 + \|B\|_2^2\right),$$

which gives the first entry. This proves Equation (63). □

4.6. **Exact Schmidt number and witnesses.** For $m \ge 2$, let $\Pi_-^{(m)}$ denote the orthogonal projection onto the antisymmetric subspace of $\mathbb{C}^m \otimes \mathbb{C}^m$, and let

$$\omega_-^{(m)} := \frac{\Pi_-^{(m)}}{\binom{m}{2}}$$

denote the corresponding normalized state.

The argument below invokes Lemma 2.1 directly rather than Theorem 1.2: what it needs is the bound on the top two Schmidt coefficients of a double-skew operator, not the partial-trace inequality derived from it. That $\mathrm{SN}(\omega_-^{(m)}) = 2$ is classical; the content here is the value for the tensor product.

Corollary 4.12 (Exact Schmidt number)**.** *With respect to the regrouped bipartition*

$$(A_1A_2) : (B_1B_2),$$

one has, for every $m, n \ge 2$,

$$\mathrm{SN}\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right) = 4. \tag{68}$$

Proof. Each standard antisymmetric basis vector has Schmidt rank 2, so $\omega_-^{(m)}$ admits a decomposition into pure states of Schmidt rank 2. Moreover, the antisymmetric subspace contains no nonzero product vector. Hence

$$\mathrm{SN}(\omega_-^{(m)}) = 2.$$

Tensoring Schmidt-rank-2 decompositions therefore gives

$$\mathrm{SN}\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right) \leq 4.$$

For the reverse inequality, let

$$Q := \Pi_-^{(m)} \otimes \Pi_-^{(n)},$$

where the first projection acts on A_1B_1 and the second on A_2B_2. Under inverse vectorization across $(A_1A_2) : (B_1B_2)$, every vector $\phi \in \operatorname{ran} Q$ has the form

$$\phi = \mathrm{vec}(K), \qquad K \in \mathfrak{so}(\mathbb{C}^m) \otimes \mathfrak{so}(\mathbb{C}^n).$$

Thus, for a unit vector $\phi \in \operatorname{ran} Q$, its Schmidt coefficients $s_1 \geq s_2 \geq s_3 \geq \cdots$ across the regrouped bipartition are the singular values of a Hilbert–Schmidt unit operator K in the above double-skew space. The case $m \neq n$ is included by the zero-extension argument of Remark 2.2. By that lemma,

$$s_1^2 + s_2^2 \leq \frac{1}{2}.$$

Since $s_3 \leq s_2 \leq s_1$,

$$2s_3^2 \leq s_1^2 + s_2^2 \leq \frac{1}{2}.$$

Consequently,

$$s_1^2 + s_2^2 + s_3^2 \leq \frac{3}{4}.$$

For any orthogonal projection Q and any k, projection duality gives

$$\begin{aligned}\sup_{\substack{\|z\|=1\\ \mathrm{SR}(z)\leq k}} \langle z, Qz\rangle &= \sup_{\substack{\phi\in\operatorname{ran} Q\\ \|\phi\|=1}} \sup_{\substack{\|z\|=1\\ \mathrm{SR}(z)\leq k}} |\langle z, \phi\rangle|^2 \\ &= \sup_{\substack{\phi\in\operatorname{ran} Q\\ \|\phi\|=1}} \sum_{j=1}^{k} s_j(\phi)^2;\end{aligned}$$

see [7]. Applying this identity with $k = 3$ gives

$$\sup_{\substack{\|z\|=1\\ \mathrm{SR}(z)\leq 3}} \langle z, Qz\rangle \leq \frac{3}{4}.$$

On the other hand,

$$\omega_-^{(m)} \otimes \omega_-^{(n)} = \frac{Q}{\operatorname{rank} Q},$$

and hence

$$\mathrm{Tr}\left[Q\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = 1.$$

If the state had Schmidt number at most 3, it would admit a convex decomposition into pure states of Schmidt rank at most 3 [15]; its expectation against Q would then be at most $3/4$, a contradiction. Therefore its Schmidt number is at least 4, proving Equation (68). □

Corollary 4.13 (Explicit Schmidt-number witnesses)**.** *Let*

$$Q = \Pi_-^{(m)} \otimes \Pi_-^{(n)}.$$

Then

$$W_2 := \frac{1}{2} I - Q$$

satisfies

$$\operatorname{Tr}(W_2 \sigma) \geq 0 \qquad \textit{whenever } \operatorname{SN}(\sigma) \leq 2,$$

and

$$W_3 := \frac{3}{4} I - Q$$

satisfies

$$\operatorname{Tr}(W_3 \sigma) \geq 0 \qquad \textit{whenever } \operatorname{SN}(\sigma) \leq 3.$$

Thus a negative expectation of W_2 certifies Schmidt number at least 3, while a negative expectation of W_3 certifies Schmidt number at least 4. For the state in Corollary 4.12,

$$\operatorname{Tr}\left[W_2 \left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = -\frac{1}{2},$$
$$\operatorname{Tr}\left[W_3 \left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = -\frac{1}{4}.$$

Appendix A. Formal verification

The proof of Theorem 1.2, and of every result it depends on, has been formalized in Lean 4. The development is available at `https://github.com/elazarg/rank-r`. It uses the toolchain `leanprover/lean4:v4.32.0`, and Mathlib is its only direct dependency, pinned in `lake-manifest.json` at revision `81a5d257c8e410db227a6665ed08f64fea08e997` (tag `v4.32.0`); that manifest also pins every transitive dependency. After `lake exe cache get`, the single command `lake build RankR` compiles it. The sources contain no `sorry`, no `axiom` declaration, no `native_decide` and no `set_option`, and `#print axioms` applied to the main theorem reports only `propext`, `Classical.choice` and `Quot.sound`; the file `RankR/Axioms.lean` wraps that check in `#guard_msgs` for each headline theorem, so it is verified by the build rather than asserted here.

The formal counterpart of Theorem 1.2 is `rank_r_partial_trace`, and it has *no* hypotheses. The Autonne–Takagi factorization is itself proved, as `Matrix.exists_takagi`; `Takagi.lean` derives Lemma 2.1 from it, and `Equivalence.lean` supplies the universal upper bound $\beta_2(\Lambda_U \otimes \Lambda_V) \leq 1$. `Sharp.lean` proves equality when both factors contain two distinct basis labels; in the degenerate dimensions the map is zero and the sharp constant is zero. `Results.lean` composes the upper-bound chain, so the development is proved end to end. The single quantitative input is that upper bound: the development's only route from Section 2 to Section 3 is Definition 2.3. Beyond Equation (3) itself the development covers the strict form of Equation (46), hence the assertion that for nonzero C equality in Equation (3) forces $\operatorname{rank} C = r$; all of Remark 1.3 — the extremizer, its rank, the equality it attains, and the optimality of both coefficients; and Lemma 4.7.

Both directions of Remark 2.2 are formalized, so [4, Thm. 2.4] is machine-checked as well, as `fgpBound_holds`. Two things should still be stated precisely. First, the mathematics of Section 2 is theirs; the formal development follows their §2 and replaces their Theorem 2.4 by Autonne–Takagi, which removes their appeals to a max–min exchange and to [7], but it is not an independent derivation. Second, what is proved is `FGPBound` as Section 2 renders it, in the homogeneous form over $|u_1\rangle\langle v_1| + |u_2\rangle\langle v_2|$ rather than over unit vectors of Schmidt rank at most two; by Equation (1) these are the same statement, but that identification is a reading of the definitions rather than a theorem of the development.

The theorems `rank_r_partial_trace_of_FGP_square` and its variants are retained in the development as a bridge to the published estimate: they carry the hypothesis `FGPBound (Fin d) (Fin` `d)` together with card $U \le d$ and card $V \le d$, and record that Fu–Gao–Park's statement, in the square form they publish, yields the same conclusion. They are no longer on the critical path. The correspondence with the results of the main text is as follows.

Manuscript	Lean
Theorem 1.2	`rank_r_partial_trace`
Autonne–Takagi, [5, Cor. 4.4.4(c)]	`Matrix.exists_takagi`
Lemma 2.1	`doubleSkewBound_of_takagi`
Equation (6)	`hsInner_symOuter_flipU`
Equation (7)	`hsInner_flipU_pairMat_nonneg`
Equation (8)	`qform_Qm_vec_pairMat_le`
Equation (9)	`sq_add_sq_le_of_takagi`
weights in step 4 of Lemma 2.1	`exists_pair_sum_weighted_le`
Remark 2.2, forward	`doubleSkewBound_of_FGP`
Remark 2.2, converse	`FGPBound_of_doubleSkewBoundQm`
[4, Thm. 2.4]	`fgpBound_holds`
zero extension in Remark 2.2	`FGPBound_of_card_le`
Definition 2.3, $J(\Phi)$	`choiOf`
Definition 2.3, β_2	`ChoiTwoBound`
Definition 2.3, attainment	`ChoiTwoAttained`, `le_of_choiTwoAttained`
Equation (12)	`choiOf_skewKraus`, `choiTwoBound_skewKraus`
Equation (12), as an equality	`choiTwoBound_skewKraus_iff`
Theorem 1.1	`thetaPositive_of_choiTwoBound`
Equation (31), Equation (32)	`norm_sq_pair_le_of_choiTwoBound`
Equation (15)	`qform_krausQ_Pneg_le`
Equation (16)	`qform_krausQ_ptransposeUV_ge`
Equation (36)	`Psi_eq_tensor_square`
$\Delta_{U\otimes V} = \Delta_U \otimes \Delta_V$	`RU_zero_RV_zero`
Equation (35)	`Lam4_transpose`
Proposition 2.5	`operatorIneq_of_choiTwoBound`, `operatorIneq_of_FGP`
Remark 2.7, r-block positivity	`isBlockPositive_psiChoi`
Equation (19)	`Ppos_sub_Pneg`
Equation (22)	`Phi4_apply`
Equation (23)	`Phi4_ptransposeUV_rankOne`
Equation (24)	`HopScaled_eq`
Equation (29)	`Tsyn_eq_delta_vtx`
Equation (33)	`norm_Tsyn_sq_le`
Equation (34)	`inner_delta_placeQ_kron_zetaV`
Equation (34), converse	`inner_delta_placeQ_zetaV_iff`
Remark 2.6, odd degree	`inner_delta_placeQ_zetaV_of_isSkew`, `not_qform_krausQ_Pneg_le_skewFam`
Lemma 3.1	`contraction_norm`, `contraction_trace`, `hsNormSq_ptraceU_rankFactor`, `hsNormSq_ptraceV_rankFactor`
Equation (43)	`exists_rankFactor`

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Manuscript	Lean
Equation (46)	`rank_mono`
strict case of Equation (46)	`rank_mono_strict`
Equation (2)	`rank_r_partial_trace_exact`
Equation (47)	`rank_r_partial_trace_exact_of_betaTwo`
strictness for rank $C < r$	`rank_r_partial_trace_strict`
equality forces rank $C = r$	`rank_eq_of_eq_rank_r_partial_trace`
trace–rank bound in Section 3	`normSq_trace_le`
Lemma 4.7	`hsNormSq_ptraceU_le`, `hsNormSq_ptraceV_le`
Equation (58)	`ptraceU_rankOne`, `ptraceV_rankOne`
extremizer $P_r \otimes \|v\rangle\langle w\|$	`projWit`
its rank and its four norms	`rank_projWit`, `hsNormSq_projWit`, `trace_projWit`, `hsNormSq_ptraceU_projWit`, `hsNormSq_ptraceV_projWit`
equality at the extremizer	`projWit_bound_eq`
$J(\Psi_r)$ attained at it	`re_qform_psiChoi_projWit`
optimality of $a = r$	`le_coeff_hsNormSq_of_bound`
optimality of $b = 1/r$	`inv_le_coeff_trace_of_bound`
Proposition 4.1, lower branches	`twoCopyScore_lower_of_le_inv`, `twoCopyScore_lower_of_inv_le`
Proposition 4.1, attaining matrices	`twoCopyScore_projWit_same`, `twoCopyScore_projWit_orthogonal`
Corollary 4.5, score condition for $r \le d$	`twoCopyNonnegative_iff`
Corollary 4.3, score identity	`twoCopyScore_adjacent_endpoint`
Remark 4.4	`twoCopyScore_strict_separation_pos`, `twoCopyScore_adjacent_strict_neg`
Theorem 4.8, four lower branches	`asymmetricScore_lower_left_of_le_inv`, `asymmetricScore_lower_left_of_inv_le`, `asymmetricScore_lower_right_of_le_inv`, `asymmetricScore_lower_right_of_inv_le`
Theorem 4.8, attaining matrices	`asymmetricScore_projWit_same`, `asymmetricScore_projWit_orthogonal`, `asymmetricScore_swappedProjWit_same`, `asymmetricScore_swappedProjWit_orthogonal`
Corollary 4.9, score condition for $r \le d$	`asymmetricNonnegative_iff`
explicit score Choi quadratic forms	`re_qform_twoCopyScoreOperator`, `re_qform_asymmetricScoreOperator`
score block positivity for $r \le d$	`isBlockPositive_twoCopyScoreOperator_iff_le_inv`,

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Manuscript	Lean
	`isBlockPositive_asymmetricScoreOperator_iff_max_le_inv`
Choi matrix of the reduction pencil	`mapChoi_reductionMap`
displayed regrouped product Choi matrix	`productReductionChoi_eq_regroup_mapChoi`,
	`productReductionChoi_eq_asymmetricScoreOperator`
positive-semidefinite Choi continuation	`reductionChoi_posSemidef`,
	`productReductionChoi_posSemidef`
product-Choi block positivity for every positive r	`isBlockPositive_productReductionChoi_self_iff_le_inv_min`, `isBlockPositive_productReductionChoi_iff_max_le_inv_min`
finite-coordinate map/Choi equivalence	`isMapRPositive_mapOfChoi_iff_isBlockPositive`
tensor product of the reduction maps	`productReductionMapLinear_apply`
map r-positivity classifications	`isMapRPositive_productReductionMapLinear_self_iff_le_inv_min`, `isMapRPositive_productReductionMapLinear_iff_max_le_inv_min`
Equation (67)	`centeredPartialTrace_le`

Three respects in which the formal statements are renderings rather than transcriptions should be recorded. First, `FGPBound` — used only for the bridge theorems above — is stated in the homogeneous form Equation (10) rather than for unit vectors, and quantifies over expressions $|u_1\rangle\langle v_1| + |u_2\rangle\langle v_2|$ rather than over vectors of Schmidt rank at most two; by Equation (1) these are the same statement. Second, the formal analogues of H_r, Φ and $P_\pm$ carry positive integer scalings, chosen so that no denominators occur; this does not affect positivity. Third, what is certified is the deduction, not the faithfulness of the formal definitions to the informal ones; the conventions those definitions encode are exactly the ones fixed in the introduction.

Constants pinned, not merely bounded. `ChoiTwoBound` is an upper bound, and monotone in its constant: it holds at 10^6 whenever it holds at all. On its own, therefore, it names no number, and the theorems stated only with `ChoiTwoBound` certify Equation (12) only as an inequality. `Sharp.lean` supplies the other half. It adds the dual predicate `ChoiTwoAttained` — the bound met with equality at some nonzero operator of rank at most two — together with the leastness principle the pair generates, `le_of_choiTwoAttained`, and then exhibits an extremizer: $M = NP$ with $N = J_U \otimes J_V$ an elementary tensor of rotation generators and P a rank-two projection, so that N being unitary makes the two singular values of M coincide, which is the equality case of the Ky Fan estimate behind $\|Q_-\|_{S(2)} = \frac{1}{2}$. When each factor has two distinct basis labels, the conclusion is `choiTwoBound_skewKraus_iff`: the constants valid for $\Lambda_U \otimes \Lambda_V$ are exactly the reals ≥ 4 in the Φ_4 scaling, so Equation (12) is an equality. If a factor has cardinality below two, the family vanishes and the sharp constant is zero. `re_qform_psiChoi_projWit` does the same for the conclusion — the r-block positivity of $J(\Psi_r)$ is attained at the extremizer of Remark 1.3, hence cannot be sharpened to a strict inequality.

A second instance of the mechanism. The edge-Kraus family $A_e = (E_{ij} - E_{ji}) \otimes (E_{01} - E_{10})$ attached to the edges of a graph is formalized in `GraphFamily.lean`, with $\beta_2(\Phi_G) = 1$ exactly (`choiTwoBound_graphKraus_iff`), and `Theta.lean` together with `ThetaBound.lean` pin its amplification threshold: for a graph containing an R-clique, $\Theta^{\Phi_G}_{R,\lambda}$ is R-positive precisely for $\lambda \geq R-1$ (`thetaPositive_graphKraus_iff`). The lower half is the witness $C = P_S \otimes E_{01}$, which is the extremizer `projWit` of Remark 1.3 read at a two-element second factor; an edge contributes -2 to the Choi pairing exactly when both of its endpoints lie in S, and the general form of that count, `two_mul_card_le_of_thetaPositive`, is an induced-average-degree obstruction. These are the development's only statements about the *optimal* amplification coefficient rather than about a valid one, and they show the bound $(r-1)\beta_2$ of Theorem 2.4 is attained.

The formalization follows the arguments of Sections 2 and 3 except at two points, where it takes a shorter route: Equation (27) is replaced by a Bessel-type duality that never forms $\mathcal{T}$ as a matrix, and the complete-graph identity of Proposition B.2 is not used, the vertex Cauchy–Schwarz estimate sufficing. Those two displays are therefore not themselves covered.

Within Section 4, `Applications.lean` covers the exact two-copy and asymmetric score curves in an infimum-free but equivalent form: each branch is a pointwise lower bound and a displayed rank-exact matrix attains it. It also covers factor interchange, the score nonnegativity thresholds for $1 \leq r \leq d$, the endpoint and strict adjacent-rank score separations, and the coordinate Choi quadratic identities. It identifies the reduction maps' Choi matrices, performs the output/input register permutation in their tensor product, and proves its block-positivity threshold for every positive r in equal local dimensions, using a positive-semidefinite Choi argument past the local dimension. `MapPos.lean` defines positivity of the actual $\mathrm{id}_{\mathbb{C}^r} \otimes \Phi$ matrix ampliation, proves the finite-coordinate Choi equivalence, identifies the product map with the tensor product of the two reduction pencils, and derives the symmetric and asymmetric map-r-positivity classifications. The development also covers the complete rank-sensitive estimate Equation (67). The Schatten-1 norm and singular-value Ky Fan duality appear nowhere in the development, so Corollary 4.10 and the final step of Corollary 4.11 remain prose-only. The mixed-state Schmidt-number arguments in Corollaries 4.12 and 4.13, and the results of Appendices B and C, are likewise not covered.

Theorem 2.4 is covered, in all four operator forms and for an arbitrary Kraus family: `Choi.lean` carries Equation (31) with (Φ, β_2) abstract and `Lifting.lean` carries Equations (13) to (16) as `qform_krausQ_Pneg_le_of_norm`, `qform_krausQ_Pneg_le`, `qform_krausQ_ptransposeUV_ge_of_norm` and `qform_krausQ_ptransposeUV_ge`, the transpose-symmetry hypothesis appearing in exactly the second and fourth. For the double-skew family, the constant remains free through the contraction layer: `qform_HopScaled_param` and `rank_r_partial_trace_exact_of_betaTwo` formalize Equation (47). This partial-trace conclusion is not a statement for an arbitrary Φ, because the contraction Equation (23) is specific to $\Lambda_U \otimes \Lambda_V$. The lifting layer is abstract, and $\Phi = \Lambda_U \otimes \Lambda_V$ enters its application through Equation (12) and that contraction identity. Theorem 1.1 is covered at general β_2, as `thetaPositive_of_choiTwoBound`: writing $\Theta^{\Phi}_{R,\lambda}$ for $\Phi \circ \tau + \lambda(\Delta_d - \frac{1}{R}\mathrm{id})$ and taking R-positivity in its Choi form, that map is R-positive at $\lambda = (R-1)\beta$ for every β admissible in Definition 2.3 and every completely positive Φ with transpose-symmetric Kraus operators. The conversion from Theorem 2.4 does not use Equation (23), which is why it survives the loss of the double-skew family: it runs Equation (16) at the *conjugated* frame $\overline{e}$ of the range factorization Equation (43) and at the test vector $\delta_{\overline{d}}$, where the negative-sector term is exactly the Choi pairing of $\Phi \circ \tau$. Both sides are double sums over the frame indices of the *bilinear* pairing $B(x,y) = \sum_w x_w y_w$ — the partial transpose is what replaces the inner product by a bilinear one — and they match term by term through $B(A^*\overline{x}, \overline{y}) = \overline{B(Ay, x)}$. Those files, and `Sectors`, `Synth`, `Frame` and `Edge` beneath them, are stated over a single space rather than over $U \otimes V$; the bipartite structure enters at the Φ_4 kernel and not before. Equation (36) is covered as `Psi_eq_tensor_square`, with the two structural inputs of Remark 2.7 — monoidality of Δ and Equation (35) — as separate lemmas, and the r-block-positivity reading of Remark 2.7 as `isBlockPositive_psiChoi`. The r-positivity statement of Theorem 2.4 is formalized in the unrolled form $\operatorname{rank} C \leq r$ rather than through a definition of

Schmidt rank, exactly as `FGPBound` is. Remark 2.6 is formalized, in both directions and past the point the remark itself argues. `inner_delta_placeQ_zetaV_iff` upgrades Equation (34) to an equivalence — orthogonality of every $(N \otimes I_Q)\zeta_{ij}$ to δ_e *is* the degree-0 hypothesis, so that hypothesis is exact and not merely sufficient — and `inner_delta_placeQ_zetaV_of_isSkew` evaluates the same pairing in odd degree, where the two terms add instead of cancelling. That the *conclusion* and not only the proof is then lost is a separate statement, and is `not_qform_krausQ_Pneg_le_skewFam`: for the one-element family $J = E_{01} - E_{10}$ on $\mathbb{C}^2$, at $r = 2$ and $y = \delta_e$, the protected right-hand side of Equation (15) vanishes identically — for every value of β_2, so no constant rescues it — while the left-hand side is 4; and the same data meets the unprotected Equation (13) with equality (`qform_krausQ_Pneg_skewFam_saturates`), so the gap between the two displays is exactly the direction δ_e. Remark 2.8 remains an observation about the mechanism: the two-dimensionality of the commutant is not formalized.

Appendix B. Gram structure and defect identities

This section records the algebraic defect structure of the rank-r proof. The main residual is represented by positive Gram operators once the range isometry is fixed; this is not claimed to be a uniform bounded-degree polynomial sum-of-squares identity in the raw Stiefel coordinates. The only literal polynomial SOS below is the elementary trace–rank residual.

For an operator C of rank at most r, define

$$\mathsf{G}_r(C) := r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \|\operatorname{Tr}_U C\|_2^2 - \|\operatorname{Tr}_V C\|_2^2, \tag{69}$$

$$\mathsf{R}_r(C) := r\|C\|_2^2 - |\operatorname{Tr} C|^2. \tag{70}$$

Proposition B.1 (Polynomial SOS for the trace–rank residual)**.** *Assume $r \le \dim(U \otimes V)$ and write*

$$C = \sum_{i=1}^{r} |e_i\rangle\langle d_i|,$$

where $e_1, \dots, e_r$ are orthonormal and zero columns $d_i = 0$ are allowed. In $(U \otimes V) \otimes \mathbb{C}^r$, put

$$\mathbf{e} := \sum_i e_i \otimes q_i, \qquad \mathbf{d} := \sum_i d_i \otimes q_i.$$

Equip the exterior square with the convention that $u_\alpha \wedge u_\beta$ for $\alpha < \beta$ is orthonormal whenever (u_α) is an orthonormal basis. Then

$$\mathsf{R}_r(C) = \|\mathbf{e} \wedge \mathbf{d}\|^2. \tag{71}$$

Equivalently, in any orthonormal coordinate system,

$$\mathsf{R}_r(C) = \sum_{\alpha<\beta} |\mathbf{e}_\alpha \mathbf{d}_\beta - \mathbf{e}_\beta \mathbf{d}_\alpha|^2. \tag{72}$$

Proof. One has

$$\|\mathbf{e}\|^2 = r, \qquad \|\mathbf{d}\|^2 = \|C\|_2^2, \qquad |\langle \mathbf{e}, \mathbf{d}\rangle| = |\operatorname{Tr} C|.$$

The complex Lagrange identity gives both formulas. □

For the remainder of this section, fix an exact rank-r factorization

$$C = \sum_{i=1}^{r} |e_i\rangle\langle d_i|, \qquad \delta_e = \sum_i e_i \otimes q_i, \qquad \delta_d = \sum_i d_i \otimes q_i,$$

and retain ρ_0, Π_e, $P_\pm$, and Φ from the proof of Proposition 2.5. Let $\mathcal{T}_- := \mathcal{T}$ be the negative-sector synthesis map in Equation (26). Choose an orthonormal basis of $\operatorname{ran} P_+$ and let $\mathcal{T}_+$ be the synthesis map whose columns are all Kraus images under Φ of those positive-sector basis vectors. Then

$$\Phi(P_\pm) = \mathcal{T}_\pm \mathcal{T}_\pm^*.$$

For a coefficient array $c = (c^{(ij)}_{\alpha\beta})$, define

$$K_{ij} := \sum_{\alpha,\beta} c^{(ij)}_{\alpha\beta} L_\alpha \otimes M_\beta.$$

At a vertex k, associate to an incident edge the signed vectors

$$v_{k,ik} := K_{ik}\bar{e}_i \quad (i < k), \qquad v_{k,kj} := -K_{kj}\bar{e}_j \quad (k < j).$$

Proposition B.2 (Complete-graph defect certificate)**.** *With the skew Kraus normalization used in Equation (28),*

$$\boxed{\begin{aligned}&(r-1)\|c\|_2^2 - \|\mathcal{T}_- c\|^2\\ &\quad = \frac{r-1}{2}\sum_{i<j}\left[\frac{1}{2}\|K_{ij}\|_2^2 - \|K_{ij}\bar{e}_i\|^2 - \|K_{ij}\bar{e}_j\|^2\right]\\ &\qquad + \frac{1}{2}\sum_{k=1}^{r}\sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e} - v_{k,f}\|^2.\end{aligned}} \tag{73}$$

Consequently $\|\mathcal{T}_-\|^2 \le r-1$ *and*

$$(r-1)(I - \Pi_e) - \mathcal{T}_-\mathcal{T}_-^* \succeq 0.$$

Proof. At every vertex use the exact variance identity

$$(r-1)\sum_{e\ni k}\|v_{k,e}\|^2 - \|\sum_{e\ni k} v_{k,e}\|^2 = \sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e} - v_{k,f}\|^2.$$

By Equations (28) and (29),

$$\begin{aligned}&(r-1)\|c\|_2^2 - \|\mathcal{T}_- c\|^2\\ &\quad = \frac{r-1}{4}\sum_{i<j}\|K_{ij}\|_2^2 - \frac{1}{2}\sum_{k=1}^{r}\|\sum_{e\ni k} v_{k,e}\|^2\\ &\quad = \frac{r-1}{4}\sum_{i<j}\|K_{ij}\|_2^2 - \frac{r-1}{2}\sum_{i<j}\left(\|K_{ij}\bar{e}_i\|^2 + \|K_{ij}\bar{e}_j\|^2\right)\\ &\qquad + \frac{1}{2}\sum_{k=1}^{r}\sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e} - v_{k,f}\|^2.\end{aligned}$$

Factoring the first two sums gives exactly Equation (73). The vertex-variance terms are nonnegative, and each edge bracket is nonnegative by Lemma 2.1. Hence $\|\mathcal{T}_-\|^2 \le r-1$. Finally, Equation (34) gives $\operatorname{ran}\mathcal{T}_- \subseteq \delta_e^\perp$, whose projection is $I - \Pi_e$, so the operator inequality follows. ☐

Proposition B.3 (Exact positive/negative-sector Gram identity)**.** *Put*

$$B_E := (r-1)(I - \Pi_e) - \mathcal{T}_-\mathcal{T}_-^*. \tag{74}$$

Then

$$\boxed{\mathsf{G}_r(C) = \|\mathcal{T}_+^*\delta_d\|^2 + \langle\delta_d, B_E\delta_d\rangle.} \tag{75}$$

Moreover, $B_E \succeq 0$.

Proof. By Lemma 3.1, $\mathsf{G}_r(C) = \langle\delta_d, H_r^{(0)}\delta_d\rangle$. Substituting Equation (24) and the two synthesis factorizations gives Equation (75). Positivity of B_E is the operator conclusion of Proposition B.2. ☐

Remark B.4 (Scope of the Gram certificate)*.* The complete-graph identity separates the global synthesis-map defect into vertex variances and edgewise double-skew deficits. Edgewise nonnegativity is precisely the Fu–Gao–Park projection estimate used in Lemma 2.1. No uniform polynomial SOS certificate in the raw range-isometry variables is claimed.

Appendix C. Generic consequences of r-block positivity

The two results collected here are separated from Section 4 because neither uses any property of the witnesses constructed there beyond r-block positivity, and Proposition C.3 does not use Theorem 1.2 at all. They are recorded because they are what makes those witnesses usable in practice.

C.1. **Matrix-valued Schmidt-number criteria.**

Corollary C.1 (A positive-map semidefinite cut)**.** *Let $1 \le r \le d^2$, let X be a finite-dimensional Hilbert space, and put*

$$s := \min\{r, d\}, \qquad \Lambda_r := \mathcal{R}_{-1/s}^{\otimes 2}.$$

If a positive semidefinite operator $\sigma \in \mathcal{L}(X \otimes (\mathbb{C}^d)^{\otimes 2})$ has Schmidt number at most r across $X : (\mathbb{C}^d)^{\otimes 2}$, then

$$(\mathrm{id}_X \otimes \Lambda_r)(\sigma) \succeq 0. \tag{76}$$

More generally, Equation (76) holds with Λ_r replaced by $\mathcal{R}_{-a} \otimes \mathcal{R}_{-b}$ whenever $a, b \ge 0$ and

$$\max\{a, b\} \le \frac{1}{\min\{r, d\}}.$$

Proof. By Corollary 4.9, the indicated map is r-positive. Decompose σ into positive multiples of pure states of Schmidt rank at most r. For each such vector, its Schmidt support on X has dimension at most r, so positivity follows from r-positivity after restricting the identity ancilla to that support. Conjugating back into X and summing the resulting positive operators proves the assertion. □

Remark C.2 (Use as an SDP cut)*.* Whenever a semidefinite relaxation contains an explicit candidate state block σ, Equation (76) is a linear matrix inequality in the entries of that block. If those entries are affine functions of a moment matrix, applying $\mathrm{id} \otimes \Lambda_r$ produces an affine matrix pencil and therefore a valid additional semidefinite constraint under a Schmidt-number-r promise. This is a necessary condition for that promise, not in general a complete semidefinite characterization of Schmidt number at most r.

C.2. **Local aggregation of rank witnesses.**

Proposition C.3 (Local-channel aggregation)**.** *Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be finite-dimensional Hilbert spaces. For each e in a finite set E, let*

$$\Phi_e^A : \mathcal{L}(\mathcal{H}_A) \to \mathcal{L}(\mathcal{K}_{A,e}), \qquad \Phi_e^B : \mathcal{L}(\mathcal{H}_B) \to \mathcal{L}(\mathcal{K}_{B,e})$$

be quantum channels, let $W_e = W_e^* \in \mathcal{L}(\mathcal{K}_{A,e} \otimes \mathcal{K}_{B,e})$ be r-block-positive, and let $\lambda_e \ge 0$. Then

$$H := \sum_{e \in E} \lambda_e (\Phi_e^A \otimes \Phi_e^B)^*(W_e) \tag{77}$$

is r-block-positive across $\mathcal{H}_A : \mathcal{H}_B$. In particular,

$$\mathrm{Tr}(H\rho) \ge 0 \qquad \textit{whenever } \mathrm{SN}(\rho) \le r.$$

Proof. Local quantum channels do not increase Schmidt number. Indeed, write

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \qquad \mathrm{SR}(\psi_k) \le r,$$

and choose Kraus operators $A_{e,i}$ and $B_{e,j}$ for Φ_e^A and Φ_e^B. Then

$$(\Phi_e^A \otimes \Phi_e^B)(\rho) = \sum_{k,i,j} p_k |(A_{e,i} \otimes B_{e,j})\psi_k\rangle\langle(A_{e,i} \otimes B_{e,j})\psi_k|,$$

and every vector in this decomposition has Schmidt rank at most r. Hence

$$\mathrm{SN}\big((\Phi_e^A \otimes \Phi_e^B)(\rho)\big) \le r$$

for every e. Therefore

$$\begin{aligned}\mathrm{Tr}(H\rho) &= \sum_{e\in E} \lambda_e \,\mathrm{Tr}\left[W_e(\Phi_e^A \otimes \Phi_e^B)(\rho)\right] \\ &\ge 0.\end{aligned}$$

□

Corollary C.4 (A local-Hamiltonian Schmidt-number certificate)**.** *In Proposition C.3, suppose each local output contains two matched d_e-dimensional pairs, with $r \le d_e$, and take*

$$W_e = \left(I - \frac{1}{r}|\Omega_e\rangle\langle\Omega_e|\right)^{\otimes 2}.$$

Then the operator H in Equation (77) satisfies

$$\mathrm{Tr}(H\rho) < 0 \quad \Longrightarrow \quad \mathrm{SN}(\rho) \ge r+1.$$

For literal subsystem tests, the channels Φ_e^A and Φ_e^B may be taken to be the corresponding partial traces, so H is a sum of local terms embedded into the global Hilbert space. More generally, any geometric locality of H is inherited from the subsystem supports of the chosen channels; arbitrary channels need not produce local Hamiltonian terms.

Remark C.5 (Positive-semidefinite pulled-back terms)*.* Write

$$\Psi_e := \Phi_e^A \otimes \Phi_e^B$$

and shift the output-space witness to

$$h_e^{\mathrm{out}} := W_e - \lambda_{\min}(W_e) I_e \succeq 0,$$

where I_e is the identity on $\mathcal{K}_{A,e} \otimes \mathcal{K}_{B,e}$. Its global pullback

$$\widetilde{h}_e := \Psi_e^*(h_e^{\mathrm{out}}) \succeq 0$$

is well defined even when the output spaces depend on e. Since Ψ_e is trace-preserving, $\Psi_e^*(I_e) = I$, and hence

$$\sum_e \lambda_e \widetilde{h}_e = H - \left(\sum_e \lambda_e \lambda_{\min}(W_e)\right) I.$$

Consequently, every normalized ρ with $\mathrm{SN}(\rho) \le r$ satisfies

$$\mathrm{Tr}\left[\sum_e \lambda_e \widetilde{h}_e \, \rho\right] \ge -\sum_e \lambda_e \lambda_{\min}(W_e).$$

For the witnesses in Corollary C.4,

$$\lambda_{\min}(W_e) = 1 - \frac{d_e}{r},$$

so the known energy baseline is

$$\sum_e \lambda_e \left(\frac{d_e}{r} - 1 \right).$$

This construction certifies Schmidt number across the prescribed bipartition; by itself it is not an NLTS statement.

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